News exposure, local diversity, and U.S. public opinion on climate change: A model-based analysis using county-level covariates

Online Appendix

In the Online Appendix we

- discuss the county characteristics,
- explain how we link climate disasters and climate-related policy events to the construction of the number of news items and their objective importance,
- show the derivation of the subjective importance function and the plausible values for the hyperparameters that determine political color and subjective importance of national television stations,
- show the approximation procedure for local power balance and the plausible values for the hyperparameters that determine county political color,
- list all candidate models and report the corresponding regression results,
- provide sensitivity analysis, and
- provide the complete set of policy experiments.

1 Introduction

This Online Appendix supplements more details for the contents that are not included in the main text due to the constraints on length and is organized as follows. In Section 2 we discuss the county characteristics. In Section 3 we discuss how to deal with objective news. In Section 4 we determine and visualize the political color of two national television stations. In Section 5 we show the derivation of subjective importance. We explain how to approximate for local power balance at two-year intervals in Section 6, and determine county political color in Section 7. We list all potential models and select our baseline model in Section 8. Sensitivity analysis is shown in Section 9. All the policies we experiment with are presented in Section 10.

2 County characteristics

We shall employ county-level political, demographic, economic, and geographic data to derive county characteristics.

Some counties have been merged or have been eliminated during this period. In order to achieve consistency and comparability, we choose the year 2020 as our benchmark, and adjust (if necessary) the data from the other years. Thus we obtain 3,111 counties and county equivalents.

We construct the following set of county characteristics:

- Voter turnout: We divide the total number of valid votes by the total population per county as a proxy to the (relative) voter turnout.
- Party share: Excluding the negligible third parties' votes, we define party share as the percentage of votes received by the democratic (or republican) party relative to the total number of votes for the two major parties.
- Population density: Population density is measured by population per square miles, that is, total population divided by land area.
- Household median and mean income: Income data have to be adjusted for inflation, and we adjusted all income-related data to 2020 dollars using the appropriate CPI-U-RS (Consumer Price Index) adjustment factor.
- Median house price: Unlike income, house prices are usually not adjusted for inflation, as a house is not only a consumption good but also an asset. House prices are typically influenced by both inflation and

other factors such as demand and the interest rate. Therefore, we use nominal house prices.

- Unemployment rate: Unemployment rate is measured as the share of the labor force without work.

- County adjacency: For each county we can identify the adjacent counties.

3 Objective news: importance, positivity, and reliability

Recall that we distinguish between three objective ingredients underlying the national news: importance, reliability, and positivity. However, in the absence of information on reliability and positivity, we assume as a default that all news items share identical, numerically neutral values for these attributes. This section only discusses the number of news and their objective importance.

3.1 Climate-related events

Anderson (2009) summarizes researches on media coverage of climate change, concluding that journalists tend to associate climate change with severe weather events, and that important policy events significantly increase the amount of news coverage. Accordingly, we link the severe U.S. climate disasters, as well as the global and national climate-related policy events to the objective importance of news coverage.

3.1.1 The U.S. climate disasters

We obtain information on U.S. billion-dollar climate disasters from the National Centers for Environmental Information (NCEI). Figure 1 shows the annual number of billion-dollar climate disasters (solid line) and their total CPI-adjusted losses (dashed line).

We then create an index $\nu_i^{d-}(\tau)$, namely current intensity of the ith disaster at time τ. This index captures the instantaneous impact of the ith disaster on news coverage. The value of $\nu_i^{d-}(\tau)$ is constructed according to the following assumptions:

- Following Houston JB, et al. (2012), who focus on major natural disasters that occurred in the U.S. from 2000 to 2010, and find that disaster

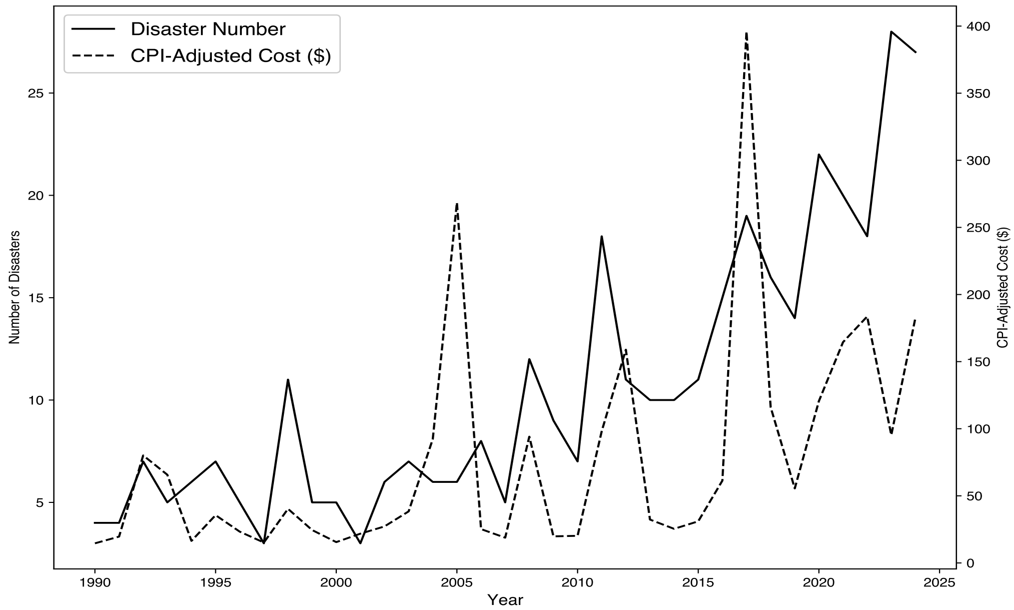


Figure 1: Number and losses of the U.S. billion-dollar disasters

news coverage lasted for 178 days (nearly 6 months) on average when excluding the extremely long news coverage of Hurricane Katrina, we assume that the effects of climate disasters last from 6 to 12 months.

- The impact of a climate disaster on news coverage gradually diminishes over month.

- The initial intensity of a climate disaster, $\nu_{i,0}^{d-}$, increases linearly with its corresponding economic loss l_i.

Since the effects of climate disasters that occurred in 1989 may have extended into 1990, we also take them into account. Consequently, there were 376 billion-dollar climate disasters in the U.S. from 1989 to 2024, with an average loss of 7.28 billion dollars (CPI-adjusted), while the maximum and minimum losses are 201.3 and 1.1 billion dollars, respectively.

Given the assumptions stated above, we define the initial intensity of the most severe disaster ($l_i = 201.3$) as 12, and that of the least severe disaster ($l_i = 1.1$) as 6. Moreover, the intensity depreciates by 1 unit per month until it reaches 0, so that the duration of impact on news coverage ranges from 6 to 12 months. For example, for the least severe climate disaster occurring at time τ, its initial intensity is $\nu_{i,0}^{d-} = 6$, and its current intensities during the affected months are:

$$\begin{array}{cccccc} \nu_i^{d-}(\tau) & \nu_i^{d-}(\tau+1) & \nu_i^{d-}(\tau+2) & \nu_i^{d-}(\tau+3) & \nu_i^{d-}(\tau+4) & \nu_i^{d-}(\tau+5) \\ 6 & 5 & 4 & 3 & 2 & 1 \end{array}$$

We further define the linear equation between initial intensity and economic loss as:

$$\nu_{i,0}^{d-} = 0.03 l_i + 6,$$

where the value of $\nu_{i,0}^{d-}$ is rounded to the nearest integer. Since $1.1 \leq l_i \leq 201.3$ (the largest disaster was Hurricane Katrina in August 2005, causing 1,833 deaths and approximately \$201.3 billion in damages), it follows (after rounding) that $6 \leq \nu_{i,0}^{d-} \leq 12$. This equation is derived based on two observed points: $(201.3, 12)$ and $(1.1, 6)$. The coefficient 0.03 and the intercept 6 are rounded to two decimal places and the nearest integer, respectively, for simplicity. Finally, the total disaster intensity at time τ is given by the sum of the current diminished intensities of all climate disasters affect that month:

$$\nu^{d-}(\tau) = \sum_i \nu_i^{d-}(\tau).$$

3.1.2 Global and national climate-related public events

In addition to climate-related disasters, we also consider major climate-related policy events like international climate conferences or legislative acts. Unlike disasters, which are always negative, policy events can be either positive or negative.

We use data from the Media and Climate Change Observatory (MeCCO), and consider the news coverage on climate change by two representative television stations: CNN and FOX. This television coverage is shown in Figure 2, and it is clear that CNN reports much more frequently on climate events than FOX. We focus on the main fluctuations and peaks, and investigate what influential climate-related policy events occurred around those points in time.

The main climate-related policy events that had serious impacts on the USA since 1990 are listed in Table 1. Since policy events typically receive much longer news coverage than disasters (Houston et al., 2012), we assume that the initial intensity of a policy event is equal to the initial intensity of the most severe disaster:

$$\nu_{i,0}^{e+} = \nu_{i,0}^{e-} = 12.$$

As before, the total policy events intensities for positive and negative events at time τ are given by the sum of the current diminished intensities

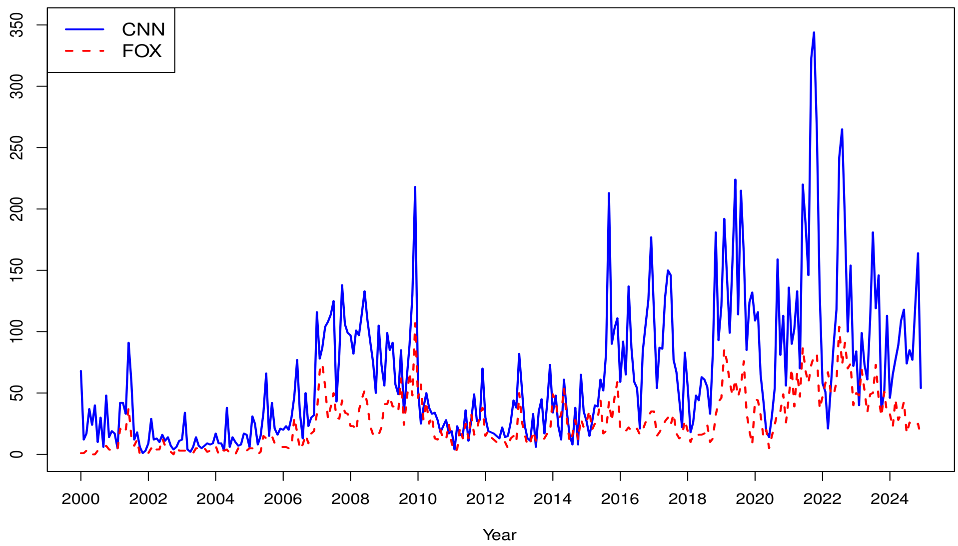


Figure 2: Monthly television news coverage of climate change in the USA

Table 1: Major climate-related events, 1990–2024

Event	Date	$\nu_{i,0}^{e+}$	$\nu_{i,0}^{e-}$
UNFCCC Adoption	1992/06	12	0
Kyoto Protocol	1997/12	12	0
Withdraw from Kyoto Protocol	2001/03	0	12
An Inconvenient Truth	2006/05	12	0
American Recovery and Reinvestment Act	2009/02	12	0
American Clean Energy and Security Act	2009/06	12	0
Copenhagen Climate Summit	2009/12	12	0
White House Office of Energy and Climate Change Policy eliminated	2011/04	0	12
Pope Francis emphasized climate change in his speech in Washington	2015/09	12	0
Paris Agreement	2015/12	12	0
Withdraw from Paris Agreement	2017/06	0	12
Global Climate Action Summit	2018/09	12	0
Green New Deal Resolution	2019/02	12	0
Rejoin Paris Agreement	2021/01	12	0
Inflation Reduction Act	2022/08	12	0

of all climate-related policy events affecting that month:

$$\nu_e^-(\tau) = \sum_i \nu_i^{e-}(\tau), \qquad \nu_e^+(\tau) = \sum_i \nu_i^{e+}(\tau).$$

3.1.3 Total climate events intensity

After aggregating the events intensities contributed by both severe disasters and policy events, we arrive at the total positive and negative climate events intensities:

$$\nu^+(\tau) = \nu_e^+(\tau), \qquad \nu^-(\tau) = \nu_d^-(\tau) + \nu_e^-(\tau),$$

which are visualized in Figure 3. Negative climate-related events are considerably more frequent than positive ones.

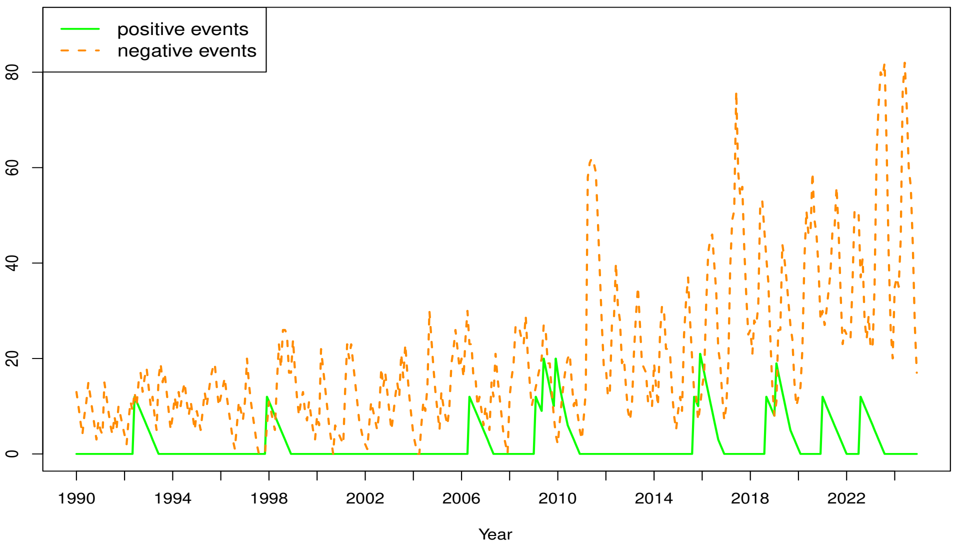


Figure 3: Monthly climate-related events intensity

3.2 Number of news items

To facilitate the transformation from climate-related objective news to news *blocks* we define the number of positive and negative news items in each month as:

$$n^+(\tau) = \left\lfloor \frac{\nu^+(\tau)}{s} \right\rfloor + 1, \qquad n^-(\tau) = \left\lfloor \frac{\nu^-(\tau)}{s} \right\rfloor + 1.$$

where $\lfloor x \rfloor$ denotes the floor function, which rounds x down to the nearest integer less than or equal to x. The additional of 1 ensures that apart from severe climate disasters and important climate-related policy events, there are still some other basic climate events with low importance, constituting the baseline climate news block. In other words, at least one positive and one negative news block exists in each month. s is a constant value to adjust the scale of news blocks number. Since $\nu^+_{max} = 21$ and $\nu^-_{max} = 82$, we set $s = 10$, so that

$$n^+_{max} = \lfloor 2.1 \rfloor + 1 = 3, \qquad n^-_{max} = \lfloor 8.2 \rfloor + 1 = 9,$$

and we have a maximum number of $q = n^+_{max} + n^-_{max} = 12$ news blocks per month.

3.3 Objective importance of each news block

We first assume that the objective importance of the most severe disaster and all climate-related policy events is equal to 1, and that of the least severe disaster is equal to 0.4. Furthermore, the objective importance of underlying baseline climate events is set at 0.2. Then similar to the definition of initial intensity, we assign importance values m^d_i and m^e_i to each disaster and policy event as follows:

$$m^d_i = 0.003 l_i + 0.4, \qquad m^e_i = 1,$$

where the value of m^d_i is rounded to one decimal place. This equation is also derived based on two observed points: $(201.3, 1)$ and $(1.1, 0.4)$. The coefficient 0.003 and the intercept 0.4 are rounded to three decimal places and one decimal place, respectively, for simplicity. Unlike intensity, the importance of each climate event remains constant over time.

Then, at time τ, the importance of each news block excluding baseline news block is determined by all severe climate disasters and policy events affecting the news coverage of that month, as well as the number of news blocks apart from the underlying baseline block $\lfloor x \rfloor$. We divide all climate-related events into $\lfloor x \rfloor$ blocks for positive and negative news items respectively, and the total climate events intensity is evenly allocated to each block:

$$\bar{U}(\tau) = \frac{\nu(\tau)}{\lfloor x \rfloor},$$

where $\bar{U}(\tau) \in \{\bar{U}^+(\tau), \bar{U}^-(\tau)\}$, $\nu(\tau) \in \{\nu^+(\tau), \nu^-(\tau)\}$.

After deriving the intensity of each block, we sort all disasters and policy events in ascending order according to their current intensities and check whether a particular event spans one or two blocks. The share of each event within a given block serves as the weight that it contributes to the objective importance of the news block. The weight of ith event contributes to kth block can be written as:

$$w_{ik}(\tau) = \frac{max\,(0, min\,(V_i(\tau), U_k(\tau)) - max\,(U_{k-1}(\tau), V_{i-1}(\tau)))}{\bar{U}(\tau)},$$

where $V_i(\tau) = \sum \nu_i(\tau)$, and $U_k(\tau) = \sum k \times \bar{U}(\tau)$. Finally, we derive the objective importance of each news block, which is a weighted average of each climate event:

$$m_k(\tau) = \sum w_{ik}(\tau) \times m_i(\tau),$$

and it is rounded to one decimal place.

3.4 An example

To give an example, we summarize all disasters and policy events that have impact on the news coverage of climate change in September 2017 based on our assumptions:

Disasters and events	Begin date	Loss
Southeast Severe Weather and Tornadoes	2017.04	1.3
South and Southeast Severe Weather	2017.04	1.2
Missouri and Arkansas Flooding	2017.04	2.2
Colorado Hail Storm	2017.05	4.3
North Central Severe Weather and Tornadoes	2017.05	1.2
Minnesota Hail Storm	2017.06	3
Midwest Severe Weather 1	2017.06	1.9
Midwest Severe Weather 2	2017.06	1.8
Withdraw form Paris Agreement	2017.06	/
Western Wildfires	2017.06	23.2
Hurricane Harvey	2017.08	160
Hurricane Irma	2017.09	64
Hurricane Maria	2017.09	115.2

Since there is no positive policy event affecting the news coverage in September 2017, the total positive climate events intensity is $\nu^+(\tau) = 0$, and thus this month only contains one baseline positive news block with 0.2 unit of objective importance. On the other hand, taking the diminishing effect into account, we obtain the total negative climate events intensity for this month $\nu^-(\tau) = 56$. Therefore, there are $n^-(\tau) = \left\lfloor \frac{56}{10} \right\rfloor + 1 = 6$ negative news

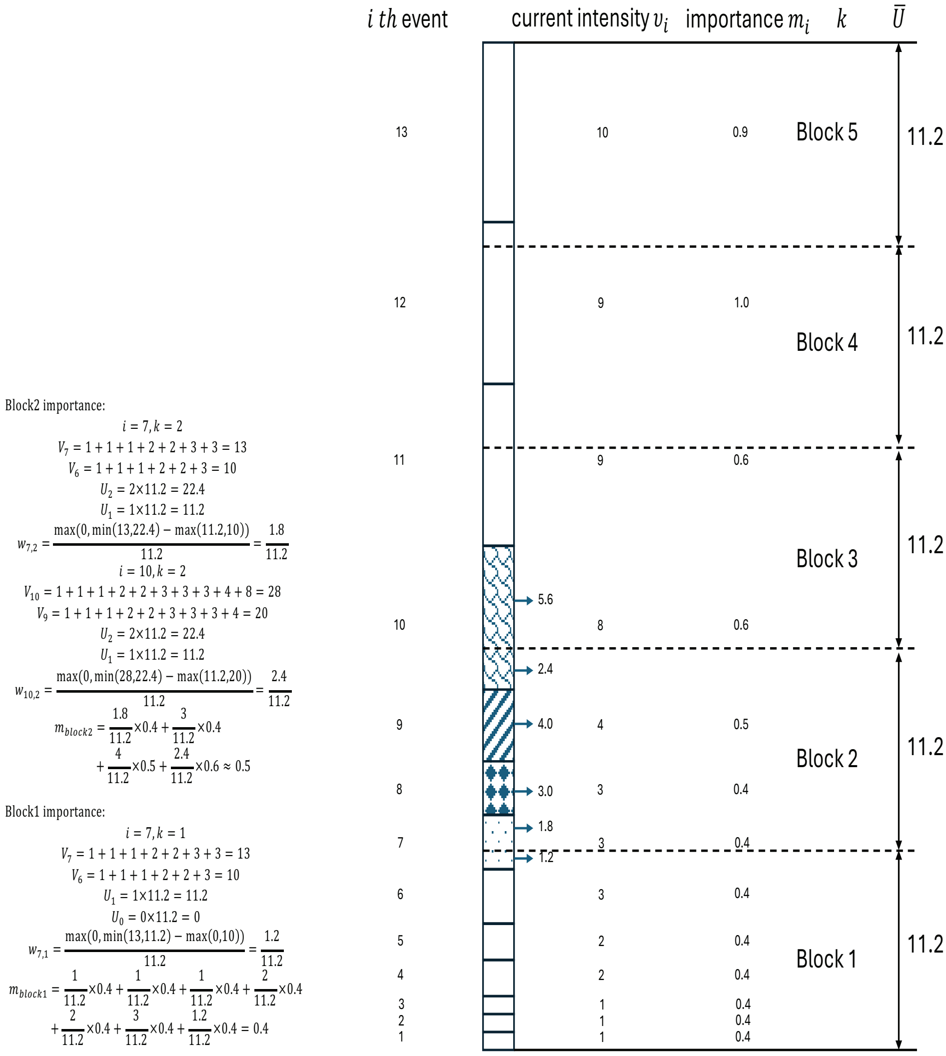


Figure 4: September 2017 negative news blocks and objective importance

block including the baseline negative news block with 0.2 unit of objective importance in that month.

Figure 4 presents the sorting and division process, and provides the specific calculations of objective importance values for negative news block 1 and 2.

Consequently, we obtain a vector with $q = 12$ elements, representing the objective importance of all news blocks in September 2017:

$$\begin{pmatrix} 0 & 0 & 0.2 & 0 & 0 & 0 & 0.2 & 0.4 & 0.5 & 0.6 & 0.9 & 0.9 \end{pmatrix}$$

The first three elements correspond to positive news, and the subsequent nine elements correspond to negative news. 0 means that the corresponding news block does not exist.

The previous discussion provides us with the (constant) importance and the (decreasing) intensity of each climate event, and we consider these pieces of information as objective facts that are made available to the two news providers. Taking the importance and intensity of each news event into account, we now bundle the news events into news blocks, where for each block we know its importance and whether it is positive or negative news (and ideally also its reliability, although we shall only use reliability in the simulations, not in the estimations due to lack of information). In our stylized world, the news providers then choose the number and the type (positive or negative) of these objective blocks depending on how much time they want to spend on climate news and whether they wish to emphasize positive or negative news.

4 National political color

The political color of two national television stations is defined as

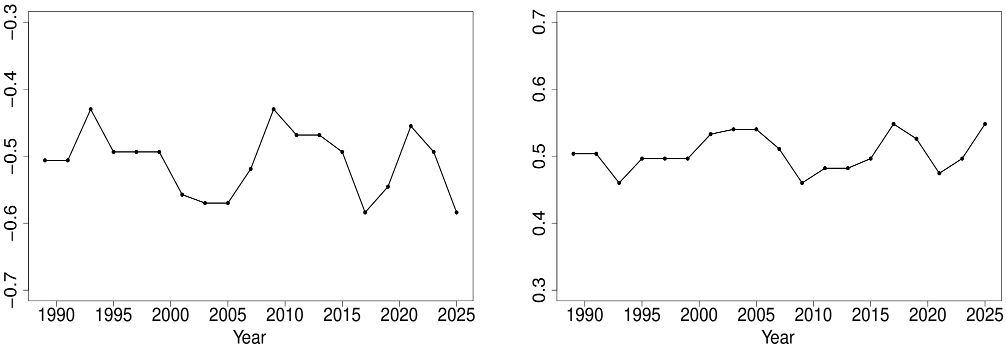
$$\bar{c}_j(t) = \beta_{0j} + \beta_{1j}\big(B(t) + \beta_2 I_T(t)\big) \qquad (j = d, r).$$

Plausible values for the parameters are

$$(\beta_{0d}, \beta_{1d}) = (-0.5, -0.07), \qquad (\beta_{0r}, \beta_{1r}) = (0.5, 0.04), \qquad \beta_2 = 0.2$$

for the left-wing and right-wing stations, respectively, but we shall also experiment with other values of the parameters. Notice that we have assumed that $|\beta_{1d}| > |\beta_{1r}|$, based on the idea that the left-wing channel responds more sharply to a change in government (both from democratic to republican and vice versa) than the right-wing channel; see Mitchell et al. (2014), Jost (2017), and Kim et al. (2022).

In Figure 5 we plot the development of the political color for the left- and right-wing stations. Given the fact that $\beta_{1d} < 0$ and $\beta_{1r} > 0$, we see that a republican government pushes the left-wing station further to the left and the right-wing station further to the right: polarization increases. This behavior is based on Dunlap and McCright (2008)—in the context of climate change—and Kim et al. (2022). The political color of the left-wing channel is not constant, because it is influenced by political power; it moves between -0.43 and -0.58. Similarly, the political power of the right-wing party moves (slightly less) between 0.46 and 0.55.



(a) democratic channel

(b) republican channel

Figure 5: Political color $\bar{c}_d(t)$ and $\bar{c}_r(t)$

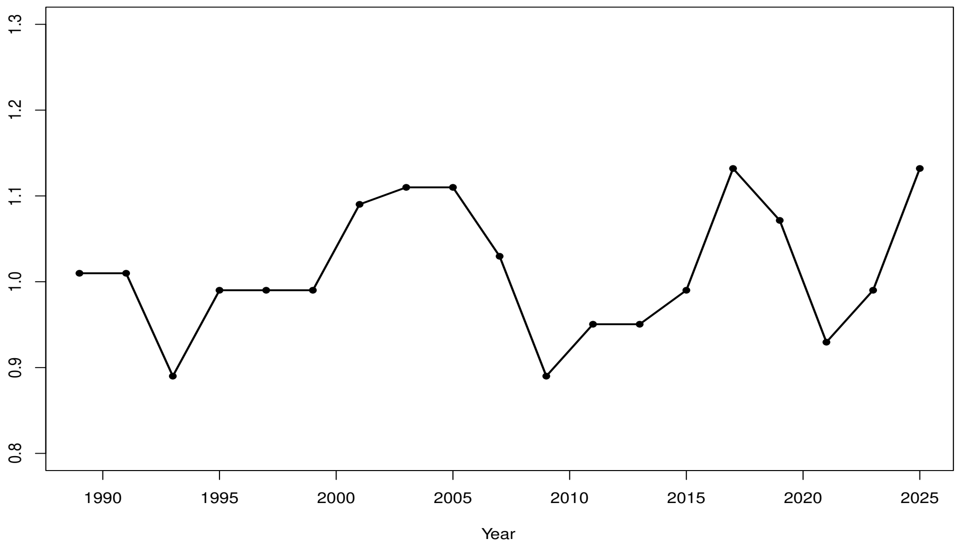


Figure 6: Political distance $\bar{c}_r(t) - \bar{c}_d(t)$ between republican and democratic channels

The increase in polarization between the two channels is plotted in Figure 6. This increase is defined as $\bar{c}_r(t) - \bar{c}_d(t)$ and, as expected, it takes a maximum in the periods 2017–2019 (first half of Trump-1) and 2025–2027 (first half of Trump-2), although polarization was also high under G. W. Bush.

5 Subjective importance

The subjective importance of the hth news item from the point of view of the jth television station is defined as a two-step process:

$$\bar{m}_j^{(h)}(\tau) = f_1\big(\bar{m}^{(h)}(\tau), \bar{c}_j(\tau)\big) \cdot f_2\big(\bar{r}^{(h)}(\tau), \bar{p}^{(h)}(\tau), \bar{c}_j(\tau)\big).$$

We specify the functions f_1 and f_2 as

$$f_1\big(\bar{m}^{(h)}(\tau), \bar{c}_j(\tau)\big) = \alpha_{1j}(\tau)\bar{m}^{(h)}(\tau)$$

and

$$f_2\big(\bar{r}^{(h)}(\tau), \bar{p}^{(h)}(\tau), \bar{c}_j(\tau)\big) = \alpha_{2j}(\tau)\bar{r}^{(h)}(\tau) + \alpha_{3j}(\tau)\bar{p}_+^{(h)}(\tau) + \alpha_{4j}(\tau)\bar{p}_-^{(h)}(\tau),$$

where we write $|\bar{p}^{(h)}| = \bar{p}_+^{(h)} + \bar{p}_-^{(h)}$ with

$$\bar{p}_+^{(h)} = \begin{cases} 0 & \text{if } \bar{p}^{(h)} \le 0, \\ |\bar{p}^{(h)}| & \text{if } \bar{p}^{(h)} > 0, \end{cases} \qquad \bar{p}_-^{(h)} = \begin{cases} |\bar{p}^{(h)}| & \text{if } \bar{p}^{(h)} \le 0, \\ 0 & \text{if } \bar{p}^{(h)} > 0. \end{cases}$$

The weights $\alpha_{1j}, \ldots, \alpha_{4j}$ depend linearly on the political color

$$\alpha_{\ell j}(\tau) = \phi_{\ell j} + \theta_{\ell j}\bar{c}_j(\tau) \qquad (\ell = 1, \ldots, 4).$$

Letting $B^*(\tau) = B(\tau) + \beta_2 I_T(\tau)$, we obtain

$$\bar{c}_d(\tau) = \beta_{0d} + \beta_{1d}B^*(\tau), \qquad \bar{c}_r(\tau) = \beta_{0r} + \beta_{1r}B^*(\tau),$$

and hence

$$\alpha_{\ell d}(\tau) = \phi_{\ell d}^* + \theta_{\ell d}^* B^*(\tau), \qquad \alpha_{\ell r}(\tau) = \phi_{\ell r}^* + \theta_{\ell r}^* B^*(\tau),$$

where

$$\begin{aligned} \phi_{\ell d}^* &= \phi_{\ell d} + \theta_{\ell d}\beta_{0d}, \qquad & \theta_{\ell d}^* &= \theta_{\ell d}\beta_{1d}, \\ \phi_{\ell r}^* &= \phi_{\ell r} + \theta_{\ell r}\beta_{0r}, \qquad & \theta_{\ell r}^* &= \theta_{\ell r}\beta_{1r}. \end{aligned}$$

This gives

$$f_1\big(\bar{m}^{(h)}(\tau), \bar{c}_j(\tau)\big) = \phi^*_{1j}\bar{m}^{(h)}(\tau) + \theta^*_{1j}\bar{m}^{(h)}(\tau)B^*(\tau),$$
$$f_2\big(\bar{r}^{(h)}(\tau), \bar{p}^{(h)}(\tau), \bar{c}_j(\tau)\big) = \Phi^{(h)}_{2j}(\tau) + \Theta^{(h)}_{2j}(\tau)B^*(\tau),$$

where

$$\Phi^{(h)}_{2j}(\tau) = \phi^*_{2j}\bar{r}^{(h)}(\tau) + \phi^*_{3j}\bar{p}^{(h)}_+(\tau) + \phi^*_{4j}\bar{p}^{(h)}_-(\tau),$$
$$\Theta^{(h)}_{2j}(\tau) = \theta^*_{2j}\bar{r}^{(h)}(\tau) + \theta^*_{3j}\bar{p}^{(h)}_+(\tau) + \theta^*_{4j}\bar{p}^{(h)}_-(\tau).$$

Consequently, we obtain

$$\frac{\bar{m}^{(h)}_j(\tau)}{\bar{m}^{(h)}(\tau)} = \delta^{(h)}_{0j}(\tau) + \delta^{(h)}_{1j}(\tau)B^*(\tau) + \delta^{(h)}_{2j}(\tau)(B^*(\tau))^2,$$

where

$$\delta^{(h)}_{0j}(\tau) = \phi^*_{1j}\Phi^{(h)}_{2j}(\tau),$$
$$\delta^{(h)}_{1j}(\tau) = \phi^*_{1j}\Theta^{(h)}_{2j}(\tau) + \Phi^{(h)}_{2j}(\tau)\theta^*_{1j},$$
$$\delta^{(h)}_{2j}(\tau) = \theta^*_{1j}\Theta^{(h)}_{2j}(\tau).$$

In order to find plausible values for the hyperparameters, we work both forwards and backwards. In the forward process we provide values for the basic parameters and then compose and judge the implied (composed) parameters; in the backward process we start with the composed parameters and work towards the basic parameters.

5.1 Forward process

We first look at extrema to obtain the bounds for the results that exhibit monotonicity. Therefore, we experiment with two news items: a positive one h_1 with $\bar{r}^{(h_1)}(\tau) = 1$, $\bar{p}^{(h_1)}_+(\tau) = 1$, $\bar{p}^{(h_1)}_-(\tau) = 0$, and another negative one h_2 with $\bar{r}^{(h_2)}(\tau) = 1$, $\bar{p}^{(h_2)}_+(\tau) = 0$, $\bar{p}^{(h_2)}_-(\tau) = 1$. The power balance $B^*(\tau)$, including Trump effect, is set from -1 to 1.2, with a step size of 0.1. After a series of experiments, a set of underlying parameters that lead to reasonable results is shown in Table 2, based on the following considerations:

- Regarding democratic media we assume that $\phi_{\ell d} > 0$, $\theta_{\ell d} < 0$, and $-1 \leq \bar{c}_d(\tau) < 0$, which implies that as the democratic media move to the left, the weights $\alpha_{\ell d}(\tau)$ increase. In contrast, regarding republican media, we assume that $\phi_{\ell r} > 0$, $\theta_{\ell r} < 0$, and $0 < \bar{c}_r(\tau) \leq 1$, so that as

Table 2: Values of the hyperparameters

β_{0d}	β_{1d}	ϕ_{1d}	θ_{1d}	ϕ_{2d}	θ_{2d}	ϕ_{3d}	θ_{3d}	ϕ_{4d}	θ_{4d}
-0.50	-0.07	0.10	-1.20	0.07	-1.10	0.03	-0.60	0.04	-0.70
β_{0r}	β_{1r}	ϕ_{1r}	θ_{1r}	ϕ_{2r}	θ_{2r}	ϕ_{3r}	θ_{3r}	ϕ_{4r}	θ_{4r}
0.50	0.04	1.10	-1.05	0.90	-0.87	0.46	-0.45	0.50	-0.48

the republican media move to the right, the weights $\alpha_{\ell r}(\tau)$ decrease. Specifically, when the national power balance moves to the right, the left-wing station becomes more left-wing and puts more weight on events related to climate change, while the right-wing station becomes more right-wing and downplays climate change related events.

- Democratic media are more sensitive to changes in political color than republican media: $|\theta_{\ell d}| > |\theta_{\ell r}|$.
- For the editors, subjective importance depends primarily on objective importance, followed by reliability, and then positivity. Moreover, for two events with same absolute positivity value, negative events attract more attention than positive events: $\phi_{1j} > \phi_{2j} > \phi_{4j} > \phi_{3j}$.

After calculation, we obtain the composed parameters of news items h_1 and h_2 from the perspective of democratic media:

$$\begin{aligned}
\phi^*_{1d} &= 0.70, & \theta^*_{1d} &= 0.084,\\
\phi^*_{2d} &= 0.62, & \theta^*_{2d} &= 0.077,\\
\phi^*_{3d} &= 0.33, & \theta^*_{3d} &= 0.042,\\
\phi^*_{4d} &= 0.39, & \theta^*_{4d} &= 0.049,\\
\Phi^{(h_1)}_{2d}(\tau) &= 0.95, & \Theta^{(h_1)}_{2d}(\tau) &= 0.119,\\
\Phi^{(h_2)}_{2d}(\tau) &= 1.01, & \Theta^{(h_2)}_{2d}(\tau) &= 0.126,
\end{aligned}$$

$$\begin{aligned}
\delta^{(h_1)}_{0d}(\tau) &= 0.665, & \delta^{(h_1)}_{1d}(\tau) &= 0.1631, & \delta^{(h_1)}_{2d}(\tau) &= 0.0100,\\
\delta^{(h_2)}_{0d}(\tau) &= 0.707, & \delta^{(h_2)}_{1d}(\tau) &= 0.1730, & \delta^{(h_2)}_{2d}(\tau) &= 0.0106.
\end{aligned}$$

$\phi^*_{1d} = 0.7$ means that in the first stage of adjustment, editor from democratic media multiplies the objective importance by 0.7 when power balance is equal to 0, and an increase of 1 unit in power balance leads to an increase of $\theta^*_{1d} = 0.084$ in the measurement.

In the second stage of adjustment, for the news item with 1 unit of positivity, we obtain:

$$\Phi_{2d}^{(h_1)}(\tau) = 0.62\bar{r}^{(h_1)}(\tau) + 0.33\bar{p}_{+}^{(h_1)}(\tau)$$
$$= \left(\frac{62}{95}\bar{r}^{(h_1)}(\tau) + \frac{33}{95}\bar{p}_{+}^{(h_1)}(\tau)\right) \times 0.95,$$

$$\Theta_{2d}^{(h_1)}(\tau) = 0.077\bar{r}^{(h_1)}(\tau) + 0.042\bar{p}_{+}^{(h_1)}(\tau),$$
$$= \left(\frac{11}{17}\bar{r}^{(h_1)}(\tau) + \frac{6}{17}\bar{p}_{+}^{(h_1)}(\tau)\right) \times 0.119.$$

According to the above equations, editor from democratic media further adjusts the importance value of positive news item in stage one by multiplying 0.95 when power balance is equal to 0, and 62/95 of the effect is provided by reliability, while 33/95 of the effect is provided by positivity. Furthermore, an increase of 1 unit in power balance leads to an increase of 0.119 in the effect, with 11/17 of the effect provided by reliability, and 6/17 of the effect provided by positivity.

Also, in the second stage of adjustment, for the news items with 1 unit of negativity, we obtain:

$$\Phi_{2d}^{(h_2)}(\tau) = 0.62\bar{r}^{(h_2)}(\tau) + 0.39\bar{p}_{-}^{(h_2)}(\tau)$$
$$= \left(\frac{62}{101}\bar{r}^{(h_2)}(\tau) + \frac{39}{101}\bar{p}_{-}^{(h_2)}(\tau)\right) \times 1.01,$$

$$\Theta_{2d}^{(h_2)}(\tau) = 0.077\bar{r}^{(h_2)}(\tau) + 0.049\bar{p}_{-}^{(h_2)}(\tau),$$
$$= \left(\frac{11}{18}\bar{r}^{(h_2)}(\tau) + \frac{7}{18}\bar{p}_{-}^{(h_2)}(\tau)\right) \times 0.126,$$

whose interpretations are analogous to those of the positive news item.

For republican media, the composed parameters of news items h_1 and h_2 are :

$$\begin{aligned}
\phi_{1r}^{*} &= 0.575, & \theta_{1r}^{*} &= -0.042,\\
\phi_{2r}^{*} &= 0.465, & \theta_{2r}^{*} &= -0.035,\\
\phi_{3r}^{*} &= 0.235, & \theta_{3r}^{*} &= -0.018,\\
\phi_{4r}^{*} &= 0.260, & \theta_{4r}^{*} &= -0.019,\\
\Phi_{2r}^{(h_1)}(\tau) &= 0.700, & \Theta_{2r}^{(h_1)}(\tau) &= -0.053,\\
\Phi_{2r}^{(h_2)}(\tau) &= 0.725, & \Theta_{2r}^{(h_2)}(\tau) &= -0.054,
\end{aligned}$$

$$\begin{aligned}
\delta_{0r}^{(h_1)}(\tau) &= 0.403, & \delta_{1r}^{(h_1)}(\tau) &= -0.0599, & \delta_{2r}^{(h_1)}(\tau) &= 0.0022,\\
\delta_{0r}^{(h_2)}(\tau) &= 0.417, & \delta_{1r}^{(h_2)}(\tau) &= -0.0615, & \delta_{2r}^{(h_2)}(\tau) &= 0.0023.
\end{aligned}$$

We can then obtain:

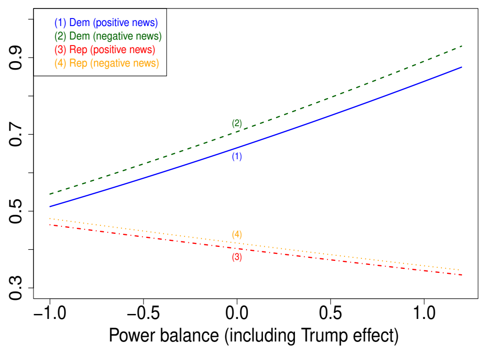
$$\Phi_{2r}^{(h_1)}(\tau) = 0.465\bar{r}^{(h_1)}(\tau) + 0.235\bar{p}_+^{(h_1)}(\tau)$$
$$= \left(\frac{93}{140}\bar{r}^{(h_1)}(\tau) + \frac{47}{140}\bar{p}_+^{(h_1)}(\tau)\right) \times 0.70,$$

$$\Theta_{2r}^{(h_1)}(\tau) = -0.035\bar{r}^{(h_1)}(\tau) + (-0.018)\bar{p}_+^{(h_1)}(\tau),$$
$$= \left(\frac{35}{53}\bar{r}^{(h_1)}(\tau) + \frac{18}{53}\bar{p}_+^{(h_1)}(\tau)\right) \times (-0.053).$$

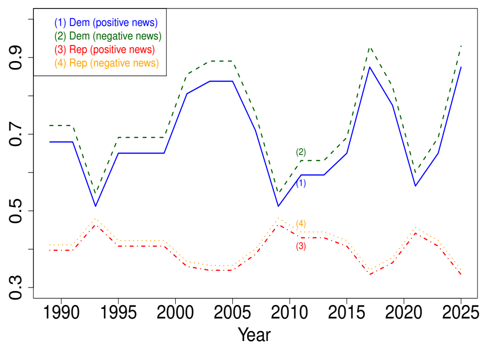
$$\Phi_{2r}^{(h_2)}(\tau) = 0.465\bar{r}^{(h_2)}(\tau) + 0.260\bar{p}_-^{(h_2)}(\tau)$$
$$= \left(\frac{93}{145}\bar{r}^{(h_2)}(\tau) + \frac{52}{145}\bar{p}_-^{(h_2)}(\tau)\right) \times 0.725,$$

$$\Theta_{2r}^{(h_2)}(\tau) = -0.035\bar{r}^{(h_2)}(\tau) + (-0.019)\bar{p}_-^{(h_2)}(\tau),$$
$$= \left(\frac{35}{54}\bar{r}^{(h_2)}(\tau) + \frac{19}{54}\bar{p}_-^{(h_2)}(\tau)\right) \times (-0.054),$$

The interpretation for republican media is similar to that for democratic media.



(a) by power balance



(b) by year

Figure 7: Ratio $\bar{m}_d/\bar{m}$ and $\bar{m}_r/\bar{m}$ of subjective to objective importance

Figure 7 illustrates the trend of the subjective importance ratio for events with 1 unit reliability and 1 unit absolute positivity, where power balance including the Trump effect is defined as $B^*(\tau) = B(\tau) + \beta_2 I_T(\tau)$, and we have taken $\bar{r}^{(h)}(\tau) = 1$, $\bar{p}_+^{(h)}(\tau) = 1$, and $\bar{p}_-^{(h)}(\tau) = -1$. Panel (a) shows that, when the power balance increases, the subjective importance of a news item increases for democratic media, but decreases for republican media.

Also, the slope of the democratic media curve is larger (in absolute value) than the slope of the republican media curve. In panel (b) we plot the annual subjective importance ratio by using the power balance at two-year intervals. The figure shows that subjective importance is sensitive to changes in the power balance in our model, that media respond stronger to negative news than to positive news, and that negative news is given relatively greater subjective importance.

5.2 Backward process

In this subsection, we begin by setting the ratio of subjective to objective importance for democratic and republican media, corresponding to the bounds of power balance. Then, we calculate the coefficients under different positions of symmetric axis.

We take the news item h_1 with 1 unit of reliability and 1 unit of positivity as an example, and assume that when national power balance is equal to -1, the ratio of subjective to objective importance is 0.5 for democratic media, and 0.5 for republican media. When national power balance is equal to 1.2, the ratio of subjective to objective importance is 0.9 for democratic media, and 0.35 for republican media. Then we have:

$$\text{democratic media:}\quad \begin{cases} 0.5 = \delta_{0d}^{(h_1)}(\tau) - \delta_{1d}^{(h_1)} + \delta_{2d}^{(h_1)}(\tau), \\ 0.9 = \delta_{0d}^{(h_1)}(\tau) + 1.2\delta_{1d}^{(h_1)} + 1.44\delta_{2d}^{(h_1)}(\tau), \\ -\frac{\delta_{1d}^{(h_1)}(\tau)}{2\delta_{2d}^{(h_1)}(\tau)} = x_d, \end{cases}$$

$$\text{republican media:}\quad \begin{cases} 0.5 = \delta_{0r}^{(h_1)}(\tau) - \delta_{1r}^{(h_1)} + \delta_{2r}^{(h_1)}(\tau), \\ 0.35 = \delta_{0r}^{(h_1)}(\tau) + 1.2\delta_{1r}^{(h_1)} + 1.44\delta_{2r}^{(h_1)}(\tau), \\ -\frac{\delta_{1r}^{(h_1)}(\tau)}{2\delta_{2r}^{(h_1)}(\tau)} = x_r, \end{cases}$$

where x_d, x_r are the value of symmetric axes needed to be experimented. The solutions of these system of equations are:

$$\text{democratic media:}\quad \begin{cases} \delta_{2d}^{(h)}(\tau) = -\frac{0.4}{4.4x_d - 0.44}, \\ \delta_{1d}^{(h)}(\tau) = -2x_d\delta_{2d}^{(h)}(\tau), \\ \delta_{0d}^{(h)}(\tau) = 0.5 - (2x_d + 1)\delta_{2d}^{(h)}(\tau), \end{cases}$$

republican media: $$\begin{cases} \delta_{2r}^{(h)}(\tau) = -\frac{0.15}{4.4x_r - 0.44}, \\ \delta_{1r}^{(h)}(\tau) = -2x_r\delta_{2r}^{(h)}(\tau), \\ \delta_{0r}^{(h)}(\tau) = 0.5 - (2x_r+1)\delta_{2r}^{(h)}(\tau). \end{cases}$$

The experiment results are shown in Table 3.

Table 3: Coefficients under different x_d and x_r

democratic media				republican media			
x_d	$\delta_{2d}^{(h)}(\tau)$	$\delta_{1d}^{(h)}(\tau)$	$\delta_{0d}^{(h)}(\tau)$	x_r	$\delta_{2r}^{(h)}(\tau)$	$\delta_{1r}^{(h)}(\tau)$	$\delta_{0r}^{(h)}(\tau)$
−1	0.08264	0.16529	0.58264	1	0.03788	−0.07576	0.38636
−2	0.04329	0.17316	0.62987	2	0.01794	−0.07177	0.41029
−3	0.02933	0.17595	0.64663	3	0.01176	−0.07053	0.41771
−4	0.02217	0.17738	0.65521	4	0.00874	−0.06993	0.42133
−5	0.01783	0.17825	0.66043	5	0.00696	−0.06957	0.42347
−6	0.01490	0.17884	0.66393	6	0.00578	−0.06934	0.42488
−7	0.01280	0.17926	0.66645	7	0.00494	−0.06917	0.42589
−8	*0.01122*	*0.17957*	*0.66835*	8	0.00432	−0.06904	0.42664
−9	0.00999	0.17982	0.66983	9	0.00383	−0.06895	0.42722
−10	0.00900	0.18002	0.67102	10	0.00344	−0.06887	0.42769
−11	0.00819	0.18018	0.67199	11	0.00313	−0.06881	0.42807
−12	0.00751	0.18032	0.67280	12	0.00286	−0.06875	0.42838
−13	0.00694	0.18043	0.67349	13	0.00264	−0.06871	0.42865
−14	0.00645	0.18053	0.67408	*14*	*0.00245*	*−0.06867*	*0.42888*
−15	0.00602	0.18061	0.67459	15	0.00229	−0.06864	0.42907
−16	0.00565	0.18069	0.67504	16	0.00214	−0.06861	0.42925
−17	0.00532	0.18075	0.67544	17	0.00202	−0.06859	0.42940
−18	0.00502	0.18081	0.67579	18	0.00190	−0.06856	0.42953
−19	0.00476	0.18087	0.67611	19	0.00180	−0.06854	0.42965
−20	0.00452	0.18091	0.67639	20	0.00171	−0.06852	0.42976

According to the particular equations that determine the ratio of subjective to objective importance given $\bar{r}^{(h_1)}(\tau) = 1$, $\bar{p}_{+}^{(h_1)}(\tau) = 1$, $\bar{p}_{-}^{(h_1)}(\tau) = 0$ from the perspective of democratic and republican television stations in the forward process:

$$\frac{\bar{m}_d^{(h_1)}(\tau)}{\bar{m}^{(h_1)}(\tau)} = 0.665 + 0.1631B^*(\tau) + 0.01(B^*(\tau))^2,$$

$$\frac{\bar{m}_r^{(h_1)}(\tau)}{\bar{m}^{(h_1)}(\tau)} = 0.403 - 0.0599B^*(\tau) + 0.0022(B^*(\tau))^2,$$

their symmetric axes are equal to −8.2 and 13.6, respectively. The computed coefficient values in the underlined italic rows $x_d = -8$, and $x_r = 14$ approach to the experiment results in the forward process. The consistency between forward and backward processes verifies the mathematical validity of this set of coefficients.

6 Approximation for local power balance

To define the power balance at the local (county) level, let $R_k(t)$ and $D_k(t)$ denote the number of votes in county k for the republican and democratic presidential candidate, respectively. We only observe these variables at four-year intervals, not two-year intervals, so we need to approximate the missing values, and we do so by considering the election results for the House of Representatives—these take place every two years. The election results are available at the state level and at the district level, but not at the county level. We shall use the data at the state level. (There are 435 districts associated with the 435 seats in the House. On average, one district thus contains about seven counties, but there are many counties that lie in two districts. Hence, using district data to approximate the counties is problematic.)

In order to approximate the results of the missing presidential elections per county, let d_t denote the percentage of democratic votes and $1 - d_t$ the percentage of republican votes at time t in the presidential elections (observed at times $t-1$ and $t+1$, but not at time t). Also, let e_t denote the percentage of democratic votes and $1 - e_t$ the percentage of republican votes at time t in the House of Parliament elections (observed at times $t-1$, t, and $t+1$). Let

$$\bar{d}_t = \frac{d_{t-1} + d_{t+1}}{2}, \qquad \bar{e}_t = \frac{e_{t-1} + e_{t+1}}{2},$$

both of which are observed. We approximate d_t in two different ways, labeled $d_t^{(a)}$ and $d_t^{(b)}$, in order to increase the stability of the approximation, namely as

$$\bar{e}_t d_t^{(a)} = \bar{d}_t e_t$$

and as

$$(1 - \bar{e}_t)(1 - d_t^{(b)}) = (1 - \bar{d}_t)(1 - e_t).$$

If $\bar{d}_t = \bar{e}_t$, then $d_t^{(a)} = d_t^{(b)} = e_t$, and if $e_t = \bar{e}_t$, then $d_t^{(a)} = d_t^{(b)} = \bar{d}_t$. But, in general, $d_t^{(a)}$ and $d_t^{(b)}$ will not be equal. We then let

$$\hat{d}_t = \frac{d_t^{(a)}}{d_t^{(a)} + (1 - d_t^{(b)})} = \frac{\bar{d}_t(1 - \bar{e}_t)e_t}{\bar{d}_t(1 - \bar{e}_t)e_t + (1 - \bar{d}_t)\bar{e}_t(1 - e_t)}$$

be our approximation for d_t. Unlike $d_t^{(a)}$ and $d_t^{(b)}$, $\hat{d}_t$ always lies between 0 and 1.

The local power balance in county k is defined as

$$B_k(t) = \frac{R_k(t) - D_k(t)}{R_k(t) + D_k(t)}.$$

Assuming that the same trend applies in each county within a given state, we obtain $B_k(t) = 1 - 2\hat{d}_t$, which will be constant during each two-year period and, like its national equivalent $B(t)$, it is a number between -1 and 1.

7 County political color

The political color in county k at time t is defined as

$$c_k(t) = \beta_0^* + \beta_1^* \bar{B}_k + \beta_2^* \left(B_k(t) - \bar{B}_k\right) + \beta_3^* B(t),$$

which can also be written as

$$c_k(t) = \beta_0^* + \left(\beta_1^* - \beta_2^*\right) \bar{B}_k + \beta_2^* B_k(t) + \beta_3^* B(t).$$

The following values are assumed for these hyperparameters:

$$\beta_0^* = 0.00, \quad \beta_1^* = 0.98, \quad \beta_2^* = 0.49, \quad \beta_3^* = 0.02,$$

so that county political color is determined primarily by its long-term power balance and current power balance, with only a small contribution from the national power balance.

8 Baseline model selection

The selection of baseline model is a result of sensitivity test. Table 4 presents the correlation coefficients between pairs of variables when $\gamma_0 = 0.5$. In sum, race diversity, age diversity, political diversity, population density, house median price, and education level are highly correlated (absolute value of coefficient over 0.9). To avoid suffering from multicollinearity, we include only one of these variables each time. In addition to these variables, news importance, news number, voter turnout (%), unemployment (%), and education diversity are always included. We also add education level, political color, and their cross term to explore the interaction effect.

Table 5 presents the empirical results across different models with γ_0 is fixed at 0.5. We find that the estimated coefficients of voter turnout, unemployment, and education diversity appear to be stable in terms of both sign and significance level. The positive effect of news importance is not significantly different from 0 in models (1) – (5), but it turns significant in models (6) and (7) after we include education level and political color. On the other hand, the effect of news number is small but positive in models (1) – (5), however it turns negative in models (6) and (7), although the effect is

Table 4: Correlation Matrix ($\gamma_0 = 0.5$)

Variable	newsimp	newsnum	racediv	agediv	edudiv	poldiv	turnout	popden	inc	houseprice	unem	edulevel	pc
newsimp	1	0.70	0.33	0.32	−0.32	−0.33	0.25	0.34	−0.06	0.29	0.05	0.30	0.07
newsnum	0.70	1	0.40	0.42	−0.37	−0.38	0.31	0.40	0	0.41	0.04	0.39	−0.14
racediv	0.33	0.40	1	0.98	−0.79	−0.95	0.76	0.99	0.11	0.97	−0.16	0.97	0.21
agediv	0.32	0.42	0.98	1	−0.80	−0.97	0.75	0.97	0.10	0.96	−0.21	0.97	0.15
edudiv	−0.32	−0.37	−0.79	−0.80	1	0.73	−0.45	−0.81	−0.31	−0.77	0.17	−0.85	−0.06
poldiv	−0.33	−0.38	−0.95	−0.97	0.73	1	−0.76	−0.95	−0.03	−0.90	0.24	−0.93	−0.32
turnout	0.25	0.31	0.76	0.75	−0.45	−0.76	1	0.71	−0.01	0.78	−0.12	0.75	0.24
popden	0.34	0.40	0.99	0.97	−0.81	−0.95	0.71	1	0.10	0.94	−0.14	0.96	0.20
inc	−0.06	0	0.11	0.10	−0.31	−0.03	−0.01	0.10	1	0.25	−0.45	0.26	−0.13
houseprice	0.29	0.41	0.97	0.96	−0.77	−0.90	0.78	0.94	0.25	1	−0.21	0.98	0.17
unem	0.05	0.04	−0.16	−0.21	0.17	0.24	−0.12	−0.14	−0.45	−0.21	1	−0.26	−0.22
edulevel	0.30	0.39	0.97	0.97	−0.85	−0.93	0.75	0.96	0.26	0.98	−0.26	1	0.21
pc	0.07	−0.14	0.21	0.15	−0.06	−0.32	0.24	0.20	−0.13	0.17	−0.22	0.21	1

always statistically insignificant. Income shows statistically significant and positive effect in models (1) – (4), but not in the other models. As for population density and house price, they show strong correlations with the demographic diversities and provide no additional insights. In model (6), where we introduce education level, political color, as well as their cross term, the coefficient estimates of cross term and income are not statistically significant, and thus we delete these two regressors in model (7). Finally, we define model (7) as our baseline model.

Table 6 gives the VIF value for each of variables in our baseline model (7). It shows very low multicollinearity for most variables, with the exception of education level. The VIF value for education level is equal to 10.278, which is a little larger than the conventional threshold of 10, implying the potential risk of higher multicollinearity. But as we attempt to discuss both education level and education diversity simultaneously to reveal the multiple roles of education, we just accept this slightly higher VIF value.

We next introduce a composite variable called 'low-diversity' index, which is a combination of race diversity, age diversity, education diversity, and political diversity. Column (2) of Table 7 reports the estimation results obtained by replacing education diversity with the low-diversity index. The weights assigned to the four diversity indices are based on the estimation results of Table 5, models (1) – (3). We test for other two sets of weights in columns (3) and (4). The results show that low-diversity index is significantly and negatively associated with public opinion about climate change, and that changing weights has small effect on coefficient estimates, except for education level. Although the effect of education level turns significantly negative, it is still small enough to be ignored.

Table 5: Comparison of seven models for $\gamma_0 = 0.5$

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Focus variables							
constant	-63.6^{***}	-696.5^{***}	-29.6^{***}	-94.1^{***}	-27.4^{***}	-58.4^{***}	-65.3^{***}
	(10.9)	(119.8)	(8.5)	(12.0)	(9.5)	(12.2)	(11.6)
news importance	0.55	0.77	0.36	0.41	0.59	1.49^{**}	1.59^{***}
	(0.55)	(0.56)	(0.56)	(0.53)	(0.57)	(0.56)	(0.56)
news number	0.15	0.07	0.20	0.17	0.18	-0.16	-0.19
	(0.16)	(0.16)	(0.16)	(0.15)	(0.16)	(0.17)	(0.16)
voter turnout	0.18^{***}	0.20^{***}	0.23^{***}	0.18^{***}	0.28^{***}	0.20^{***}	0.23^{***}
	(0.04)	(0.04)	(0.04)	(0.03)	(0.04)	(0.05)	(0.04)
median income	0.76^{*}	1.02^{**}	1.12^{***}	1.05^{***}	0.16	-0.50	
	(0.40)	(0.41)	(0.43)	(0.39)	(0.41)	(0.40)	
unemployment	-0.75^{***}	-0.69^{***}	-0.68^{***}	-0.75^{***}	-0.77^{***}	-0.84^{***}	-0.79^{***}
	(0.06)	(0.06)	(0.06)	(0.06)	(0.06)	(0.06)	(0.05)
pop density				0.01^{***}			
				(0.00)			
house price					0.06^{**}		
					(0.03)		
Auxiliary variables: diversity							
race	33.1^{***}						
	(5.3)						
education	139.8^{***}	143.0^{***}	114.3^{***}	170.7^{***}	86.7^{***}	145.4^{***}	149.9^{***}
	(16.7)	(18.2)	(15.8)	(16.9)	(16.1)	(20.3)	(20.2)
age		681.1^{***}					
		(119.8)					
political			-21.1^{***}				
			(4.3)				
Auxiliary variables: levels							
education level						0.37^{***}	0.36^{***}
						(0.07)	(0.07)
political color						-21.4^{***}	-19.6^{***}
						(3.1)	(3.0)
edu level $\times$ political color						-0.62	
						(0.48)	
Observations	420	420	420	420	420	420	420
Res. SE	3.41	3.44	3.48	3.30	3.55	3.34	3.34

Note: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 6: VIF of Model (7) ($\gamma_0 = 0.5$)

Variable	VIF
news importance	2.158
news number	2.399
education diversity	5.353
voter turnout	3.367
unemployment	1.179
education level	10.278
political color	1.288

Table 7: Introduction of low-diversity index

race weight	2	2	3
education weight	6	4	7
age weight	5	3	5
political weight	1	1	1
Focus variables			
constant	119.4***	117.8***	130.6***
	(13.1)	(13.6)	(13.0)
news importance	1.40**	1.30**	1.34**
	(0.55)	(0.55)	(0.54)
news number	−0.14	−0.12	−0.15
	(0.16)	(0.16)	(0.16)
voter turnout	0.27***	0.30***	0.26***
	(0.04)	(0.04)	(0.04)
unemployment	−0.90***	−0.92***	−0.93***
	(0.05)	(0.06)	(0.05)
Auxiliary variables: diversity			
low-diversity index	−282.2***	−249.7***	−277.3***
	(37.1)	(34.8)	(32.7)
Auxiliary variables: levels			
education level	0.02	−0.08**	−0.11***
	(0.04)	(0.04)	(0.04)
political color	−15.6***	−14.4***	−16.4***
	(2.9)	(2.9)	(2.9)
Observations	420	420	420
Res. SE	3.33	3.36	3.28

Note: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

9 Sensitivity analysis

The baseline model depends on a set of assumptions and specific values of hyperparameters. We deviate from the baseline model in one direction at a time to conduct sensitivity analysis, and see whether the coefficient estimates for our most interested variables are affected.

County interaction.

γ_0 is set to 0.5 in the baseline model. We experiment with values from 0 to 0.9 with a step size of 0.1. According to Table 8, the coefficient estimates of voter turnout, unemployment, education diversity, education level, and political color are always statistically significant at the 1% level. The effect of news importance is always significant, while the level of significance decreases. The effect of news number is significant at 10% level only when $\gamma_0 = 0.0$. So in most cases, we cannot identify any effect of the number of news items. Noticeably, the sign of each coefficient remains unchanged, although the absolute value of all coefficients decreases with γ_0, which is not a surprise, as the effect of each variable is absorbed by spatial interactions.

Table 8: Sensitivity test – γ_0

γ_0	0.0	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
Focus variables										
constant	-154.6^{***}	-135.8^{***}	-117.4^{***}	-99.4^{***}	-82.1^{***}	-65.3^{***}	-49.3^{***}	-34.1^{***}	-20.0^{***}	-7.7^{***}
	(21.6)	(19.7)	(17.7)	(15.7)	(13.7)	(11.6)	(9.4)	(7.2)	(4.9)	(2.5)
news importance	3.41^{***}	3.05^{***}	2.69^{***}	2.33^{***}	1.96^{***}	1.59^{***}	1.23^{***}	0.87^{**}	0.53^{**}	0.22^{*}
	(1.08)	(0.97)	(0.87)	(0.77)	(0.66)	(0.56)	(0.45)	(0.34)	(0.23)	(0.12)
news number	-0.47^{*}	-0.42	-0.36	-0.30	-0.25	-0.19	-0.13	-0.08	-0.03	-0.00
	(0.28)	(0.26)	(0.24)	(0.22)	(0.19)	(0.16)	(0.14)	(0.10)	(0.07)	(0.04)
voter turnout	0.50^{***}	0.45^{***}	0.39^{***}	0.34^{***}	0.28^{***}	0.23^{***}	0.18^{***}	0.14^{***}	0.09^{***}	0.05^{***}
	(0.08)	(0.07)	(0.07)	(0.06)	(0.05)	(0.04)	(0.04)	(0.03)	(0.02)	(0.01)
unemployment	-1.62^{***}	-1.45^{***}	-1.28^{***}	-1.12^{***}	-0.95^{***}	-0.79^{***}	-0.63^{***}	-0.47^{***}	-0.31^{***}	-0.16^{***}
	(0.11)	(0.10)	(0.09)	(0.08)	(0.07)	(0.05)	(0.04)	(0.03)	(0.02)	(0.01)
Auxiliary variables: diversity										
education	336.8^{***}	298.1^{***}	260.0^{***}	222.6^{***}	185.8^{***}	149.9^{***}	115.1^{***}	81.6^{***}	49.8^{***}	20.9^{***}
	(37.3)	(34.1)	(30.7)	(27.3)	(23.8)	(20.2)	(16.4)	(12.5)	(8.5)	(4.3)
Auxiliary variables: levels										
education level	0.56^{***}	0.53^{***}	0.50^{***}	0.46^{***}	0.41^{***}	0.36^{***}	0.30^{***}	0.23^{***}	0.16^{***}	0.08^{***}
	(0.11)	(0.10)	(0.09)	(0.09)	(0.08)	(0.07)	(0.06)	(0.05)	(0.04)	(0.02)
political color	-43.4^{***}	-38.6^{***}	-33.7^{***}	-28.9^{***}	-24.2^{***}	-19.6^{***}	-15.1^{***}	-10.8^{***}	-6.6^{***}	-2.9^{***}
	(5.6)	(5.1)	(4.6)	(4.1)	(3.5)	(3.0)	(2.4)	(1.8)	(1.2)	(0.6)
Observations	420	420	420	420	420	420	420	420	420	420
Res. SE	3.23	3.25	3.27	3.29	3.32	3.34	3.37	3.41	3.45	3.51

Note: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Trump effect.
In our setup, the parameter that captures the effect of Trump administration β_2 is set to 0.2, but we could change the magnitude of it, by raising the value or decreasing the value. As Table 9 shows, the coefficient estimates respond little to changes in Trump effect.

National political color.
Based on our original setting for the political color of two national television stations, we experiment two deviations.

First, we change the long-term value of national political color, to allow the democratic and republican stations to be more (less) left or more (less) right in Table 10. Second, we have assumed that more right leaning national power balance pushes the democratic station to be more left, and pushes the republican station to be more right, thus reinforces the polarization. We adjust the sensitivity that national television stations respond to the change in national power balance, and also we change the sign for β_{1d} and β_{1r} to experiment with the opposing assumption that increasing national power balance decreases polarization, as reported in Table 11.

In Table 10, the coefficients of voter turnout, unemployment, education diversity, education level, and political color are always statistically significant at the 1% level, and their values show very small variation. In the case that republican national station becomes more right in column (5), the coefficient of news importance increases a lot, indicating that stronger polarization in media may amplify the impact of news importance. The effect of news number is insignificant and small in most cases.

In Table 11, the coefficient estimates appear to be robust to the changes in the magnitude of β_{1d} and β_{1r}. However, when the signs of β_{1d} and β_{1r} change, the positive effect of news importance decreases a lot, and the significance level also decreases.

Subjective weights of news ingredients.

We specify parameter values that determining how national television stations assign weights to the three ingredients (objective importance, reliability, positivity) of news, which in turn determine the subjective importance. We experiment with the subjective weights of importance and negativity as examples, and find that the results are very similar, as reported in Table 12 and 13.

National media thresholds.

In the baseline model, national television stations set changeable thresholds when filtering news items. We either raise or lower these thresholds to test their robustness.

Table 14 shows the estimation robustness when the thresholds of national media change. The coefficient estimates of voter turnout, unemployment, education diversity, education level, and political color are always statistically significant at the 1% level, and their values show very small variation. However, the results show that the effects of news are somewhat sensitive to the changes in national thresholds. Specifically, the magnitude of positive effect of news importance varies a lot across different set of thresholds. On the other hand, the effect of news number remains insignificant in most cases, but it becomes significantly negative when the threshold of republican media is raised in the last column.

Severe events.

The definition of severe events affects the number of non-severe events that enter the filtering process. We change the threshold for defining severe events in Table 15, and find that it makes almost no difference to the results.

News ingredients.

Since we have no information for the objective positivity and reliability, we assign identical, numerically neutral values of positivity and reliability to all news items. We experiment with different values in Table 16. Again, the coefficient estimates of voter turnout, unemployment, education diversity, education level, and political color are quite stable. However, the positive effect of news importance becomes much stronger when positivity or reliability of all news items increases. In addition, the negative effect of news number is estimated more precisely and becomes marginally stronger.

In summary, the parameter estimates of baseline model are very robust across most sensitivity tests, with the exception for news variables. The estimates of news importance and news number are somewhat sensitive to the assumptions and settings. In specific, the magnitude of the effect of news importance may change a lot, while the effect of news number turns significantly negative in some cases.

Table 9: Sensitivity test – Trump effect

β_2	0.1	0.2	0.5	0.9
Focus variables				
constant	-65.3^{***}	-65.3^{***}	-65.6^{***}	-65.9^{***}
	(11.6)	(11.6)	(11.6)	(11.6)
news importance	1.59^{***}	1.59^{***}	1.64^{***}	1.69^{***}
	(0.56)	(0.56)	(0.55)	(0.55)
news number	-0.19	-0.19	-0.21	-0.23
	(0.16)	(0.16)	(0.16)	(0.16)
voter turnout	0.23^{***}	0.23^{***}	0.23^{***}	0.23^{***}
	(0.04)	(0.04)	(0.04)	(0.04)
unemployment	-0.79^{***}	-0.79^{***}	-0.79^{***}	-0.79^{***}
	(0.05)	(0.05)	(0.05)	(0.05)
Auxiliary variables: diversity				
education	149.9^{***}	149.9^{***}	150.3^{***}	150.9^{***}
	(20.2)	(20.2)	(20.2)	(20.2)
Auxiliary variables: levels				
education level	0.36^{***}	0.36^{***}	0.36^{***}	0.36^{***}
	(0.07)	(0.07)	(0.07)	(0.07)
political color	-19.6^{***}	-19.6^{***}	-19.8^{***}	-19.8^{***}
	(3.0)	(3.0)	(3.0)	(2.9)
Observations	420	420	420	420
Res. SE	3.34	3.34	3.34	3.34

Note: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 10: Sensitivity test – level of national political color

β_{0d}	-0.5	-0.6	-0.4	-0.5	-0.5
β_{0r}	0.5	0.5	0.5	0.6	0.4
Focus variables					
constant	-65.3^{***}	-65.7^{***}	-64.8^{***}	-66.0^{***}	-65.2^{***}
	(11.6)	(11.6)	(11.6)	(11.5)	(11.6)
news importance	1.59^{***}	1.59^{***}	1.41^{**}	1.89^{***}	1.58^{***}
	(0.56)	(0.49)	(0.58)	(0.49)	(0.56)
news number	-0.19	-0.23	-0.10	-0.48^{***}	-0.10
	(0.16)	(0.16)	(0.16)	(0.18)	(0.13)
voter turnout	0.23^{***}	0.24^{***}	0.23^{***}	0.24^{***}	0.23^{***}
	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)
unemployment	-0.79^{***}	-0.79^{***}	-0.79^{***}	-0.79^{***}	-0.80^{***}
	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)
Auxiliary variables: diversity					
education	149.9^{***}	149.9^{***}	149.0^{***}	150.8^{***}	149.9^{***}
	(20.2)	(20.1)	(20.2)	(20.0)	(20.2)
Auxiliary variables: levels					
education level	0.36^{***}	0.36^{***}	0.36^{***}	0.37^{***}	0.35^{***}
	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)
political color	-19.6^{***}	-19.7^{***}	-19.1^{***}	-21.3^{***}	-19.1^{***}
	(3.0)	(2.9)	(3.0)	(3.0)	(2.9)
Observations	420	420	420	420	420
Res. SE	3.34	3.34	3.35	3.32	3.34

Note: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 11: Sensitivity test – slope of national political color

β_{1d}	-0.07	-0.09	-0.05	0.07
β_{1r}	0.04	0.05	0.03	-0.04
Focus variables				
constant	-65.3^{***}	-65.8^{***}	-65.0^{***}	-64.0^{***}
	(11.6)	(11.6)	(11.6)	(11.5)
news importance	1.59^{***}	1.67^{***}	1.58^{***}	1.03^{*}
	(0.56)	(0.56)	(0.57)	(0.59)
news number	-0.19	-0.22	-0.18	0.06
	(0.16)	(0.16)	(0.17)	(0.17)
voter turnout	0.23^{***}	0.23^{***}	0.23^{***}	0.23^{***}
	(0.04)	(0.04)	(0.04)	(0.04)
unemployment	-0.79^{***}	-0.79^{***}	-0.79^{***}	-0.79^{***}
	(0.05)	(0.05)	(0.05)	(0.06)
Auxiliary variables: diversity				
education	149.9^{***}	150.7^{***}	149.4^{***}	147.9^{***}
	(20.2)	(20.2)	(20.1)	(20.1)
Auxiliary variables: levels				
education level	0.36^{***}	0.36^{***}	0.36^{***}	0.34^{***}
	(0.07)	(0.07)	(0.07)	(0.07)
political color	-19.6^{***}	-19.7^{***}	-19.6^{***}	-18.1^{***}
	(3.0)	(2.9)	(3.0)	(2.8)
Observations	420	420	420	420
Res. SE	3.34	3.34	3.34	3.35

Note: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 12: Sensitivity test – subjective weights of importance

ϕ_{1d}	0.1	0.15	0.08	0.1	0.1
ϕ_{1r}	1.1	1.1	1.1	1.15	1.08
Focus variables					
constant	-65.3^{***}	-66.2^{***}	-65.3^{***}	-66.5^{***}	-65.0^{***}
	(11.6)	(11.6)	(11.6)	(11.5)	(11.6)
news importance	1.59^{***}	1.85^{***}	1.58^{***}	1.93^{***}	1.59^{***}
	(0.56)	(0.56)	(0.56)	(0.57)	(0.56)
news number	-0.19	-0.29^{*}	-0.18	-0.25	-0.24
	(0.16)	(0.17)	(0.16)	(0.16)	(0.16)
voter turnout	0.23^{***}	0.23^{***}	0.23^{***}	0.23^{***}	0.23^{***}
	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)
unemployment	-0.79^{***}	-0.79^{***}	-0.79^{***}	-0.80^{***}	-0.79^{***}
	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)
Auxiliary variables: diversity					
education	149.9^{***}	151.5^{***}	149.9^{***}	151.9^{***}	149.3^{***}
	(20.2)	(20.1)	(20.2)	(20.1)	(20.2)
Auxiliary variables: levels					
education level	0.36^{***}	0.37^{***}	0.36^{***}	0.36^{***}	0.36^{***}
	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)
political color	-19.6^{***}	-20.1^{***}	-19.6^{***}	-20.3^{***}	-19.9^{***}
	(3.0)	(2.9)	(3.0)	(3.0)	(3.0)
Observations	420	420	420	420	420
Res. SE	3.34	3.33	3.34	3.33	3.35

Note: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 13: Sensitivity test – subjective weights of negativity

ϕ_{4d}	0.04	0.05	0.03	0.04	0.04
ϕ_{4r}	0.50	0.50	0.50	0.51	0.49
Focus variables					
constant	-65.3^{***}	-65.2^{***}	-65.1^{***}	-65.3^{***}	-65.0^{***}
	(11.6)	(11.6)	(11.6)	(11.6)	(11.6)
news importance	1.59^{***}	1.62^{***}	1.55^{***}	1.59^{***}	1.59^{***}
	(0.56)	(0.56)	(0.56)	(0.56)	(0.56)
news number	-0.19	-0.20	-0.17	-0.19	-0.24
	(0.16)	(0.17)	(0.16)	(0.16)	(0.16)
voter turnout	0.23^{***}	0.23^{***}	0.23^{***}	0.23^{***}	0.23^{***}
	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)
unemployment	-0.79^{***}	-0.79^{***}	-0.79^{***}	-0.79^{***}	-0.79^{***}
	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)
Auxiliary variables: diversity					
education	149.9^{***}	149.8^{***}	149.6^{***}	149.9^{***}	149.3^{***}
	(20.2)	(20.1)	(20.2)	(20.2)	(20.2)
Auxiliary variables: levels					
education level	0.36^{***}	0.36^{***}	0.36^{***}	0.36^{***}	0.36^{***}
	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)
political color	-19.6^{***}	-19.6^{***}	-19.6^{***}	-19.6^{***}	-19.9^{***}
	(3.0)	(3.0)	(3.0)	(3.0)	(3.0)
Observations	420	420	420	420	420
Res. SE	3.34	3.34	3.34	3.34	3.35

Note: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 14: Sensitivity test – national media thresholds

	Deviation from mean				
democratic	− 0.03	−0.04	−0.02	−0.03	−0.03
republican	+ 0.02	+0.02	+0.02	+0.01	+0.03
Focus variables					
constant	-65.3^{***}	-64.2^{***}	-65.4^{***}	-66.3^{***}	-65.9^{***}
	(11.6)	(11.6)	(11.6)	(11.5)	(11.5)
news importance	1.59^{***}	1.25^{**}	1.40^{***}	1.93^{***}	1.91^{***}
	(0.56)	(0.57)	(0.50)	(0.56)	(0.49)
news number	−0.19	−0.05	−0.15	−0.11	-0.48^{***}
	(0.16)	(0.16)	(0.17)	(0.10)	(0.18)
voter turnout	0.23^{***}	0.23^{***}	0.23^{***}	0.23^{***}	0.24^{***}
	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)
unemployment	-0.79^{***}	-0.79^{***}	-0.79^{***}	-0.79^{***}	-0.79^{***}
	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)
Auxiliary variables: diversity					
education	149.9^{***}	148.1^{***}	149.6^{***}	151.8^{***}	150.7^{***}
	(20.2)	(20.2)	(20.2)	(20.1)	(19.9)
Auxiliary variables: levels					
education level	0.36^{***}	0.35^{***}	0.36^{***}	0.35^{***}	0.37^{***}
	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)
political color	-19.6^{***}	-18.8^{***}	-19.0^{***}	-18.9^{***}	-21.3^{***}
	(3.0)	(3.0)	(2.9)	(2.8)	(3.0)
Observations	420	420	420	420	420
Res. SE	3.34	3.35	3.34	3.33	3.32

Note: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 15: Sensitivity test – severe events

Objective importance	≥ 0.6	≥ 0.7	≥ 0.8
Focus variables			
constant	-66.4^{***}	-65.3^{***}	-66.2^{***}
	(11.6)	(11.6)	(11.6)
news importance	1.93^{***}	1.59^{***}	1.62^{***}
	(0.57)	(0.56)	(0.51)
news number	-0.29^{*}	-0.19	-0.22
	(0.16)	(0.16)	(0.16)
voter turnout	0.23^{***}	0.23^{***}	0.23^{***}
	(0.04)	(0.04)	(0.04)
unemployment	-0.79^{***}	-0.79^{***}	-0.79^{***}
	(0.05)	(0.05)	(0.05)
Auxiliary variables: diversity			
education	151.5^{***}	149.9^{***}	151.5^{***}
	(20.1)	(20.2)	(20.2)
Auxiliary variables: levels			
education level	0.37^{***}	0.36^{***}	0.36^{***}
	(0.07)	(0.07)	(0.07)
political color	-20.2^{***}	-19.6^{***}	-20.1^{***}
	(2.9)	(3.0)	(3.0)
Observations	420	420	420
Res. SE	3.33	3.34	3.34

Note: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 16: Sensitivity test – positivity and reliability

Positivity	±0.5	±0.3	±0.7	±0.5	±0.5
Reliability	0.5	0.5	0.5	0.3	0.7
Focus variables					
constant	-65.3^{***}	-64.2^{***}	-66.8^{***}	-64.2^{***}	-67.2^{***}
	(11.6)	(11.6)	(11.5)	(11.6)	(11.5)
news importance	1.59^{***}	1.39^{**}	2.05^{***}	1.59^{***}	2.04^{***}
	(0.56)	(0.57)	(0.55)	(0.56)	(0.54)
news number	-0.19	-0.21	-0.33^{**}	-0.34^{*}	-0.34^{**}
	(0.16)	(0.17)	(0.16)	(0.19)	(0.15)
voter turnout	0.23^{***}	0.23^{***}	0.24^{***}	0.23^{***}	0.24^{***}
	(0.04)	(0.04)	(0.04)	(0.04)	(0.04)
unemployment	-0.79^{***}	-0.78^{***}	-0.80^{***}	-0.78^{***}	-0.80^{***}
	(0.05)	(0.05)	(0.05)	(0.05)	(0.05)
Auxiliary variables: diversity					
education	149.9^{***}	147.9^{***}	151.7^{***}	147.7^{***}	152.4^{***}
	(20.2)	(20.2)	(20.0)	(20.1)	(20.0)
Auxiliary variables: levels					
education level	0.36^{***}	0.36^{***}	0.37^{***}	0.37^{***}	0.38^{***}
	(0.07)	(0.07)	(0.07)	(0.07)	(0.07)
political color	-19.6^{***}	-19.5^{***}	-20.9^{***}	-20.2^{***}	-20.8^{***}
	(3.0)	(2.9)	(3.0)	(3.0)	(2.9)
Observations	420	420	420	420	420
Res. SE	3.34	3.35	3.32	3.35	3.32

Note: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

10 Policy experiments

Although we do not have county-level opinion data in the estimation stage, we can now use the estimated parameters of our baseline model to predict public opinion on climate change for each county.

$$\begin{aligned} y(\tau) = A^{-1}(\gamma_0)[&-65.3X_1 + 1.59X_2(\tau-1) + (-0.19)X_3(\tau-1) \\ &+ 0.23X_4(\tau-1) + (-0.79)X_5(\tau-1) + 149.9X_6(\tau-1) \\ &+ 0.36X_7(\tau-1) + (-19.6)X_8(\tau-1)], \end{aligned}$$

where X_1, $X_2(\tau-1)$, $X_3(\tau-1)$, $X_4(\tau-1)$, $X_5(\tau-1)$, $X_6(\tau-1)$, $X_7(\tau-1)$, $X_8(\tau-1)$ refer to constant, news importance, news number, voter turnout, unemployment, education diversity, education level, and political color, respectively. We change the input of county-level data to simulate a series of policies. Moreover, since we have the information for the county-level political colors, we can further divide all the counties into two groups, democratic and republican at each time point. This helps us to distinguish different trends in public opinion between democratic and republican counties.

Investment in public education.

We consider that federal government implements a nationwide policy to

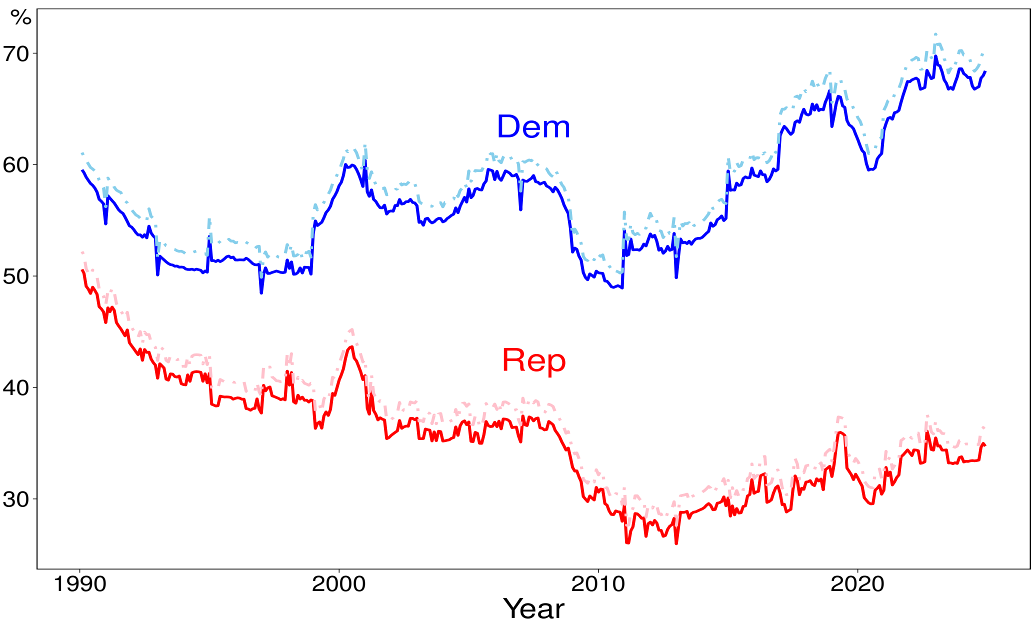


Figure 8: Education level increases by 25% quartile

improve education level of the public. However, the effects on each of the counties are not necessarily the same. Therefore, we compute the 50% and 75% quartiles of education level for each county across all months, and then assume that the nationwide policy succeeds in improving the education level of each county by its own difference between 75% and 50% quartiles. According to Figure 8, a 25% interquartile increase in the education level leads to a 1.6 percentage point (1.5 percentage point) increase in public opinion of democratic (republican) counties. But notice that in this case, the education diversity remains unchanged.

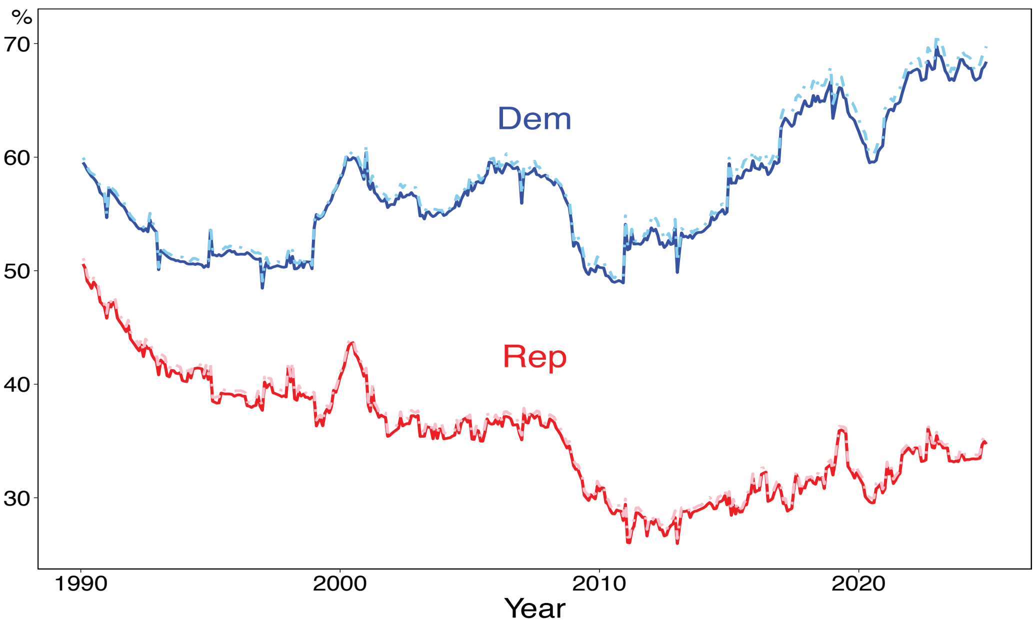


Figure 9: Education level increases by 25% quartile and education diversity decreases by 5% quantile

Next, consider a scenario where there are several levels of education, low, medium low, average, medium high, and high, and the government only invests in the average group to push it to medium high. This will increase the overall level of education, but meanwhile the education diversity declines because more people are concentrated in one group. In Figure 9, we find that even a small decline in education diversity can greatly impede the positive effect of improving education level. When education diversity decreases by the difference between 50% and 45% percentiles, the increase in public opinion declines to 0.6 percentage points (0.4 percentage points) for democratic (republican) counties.

If we further assume that the decrease in education diversity raises to the difference between 50% and 40% percentiles, the positive effect of in-

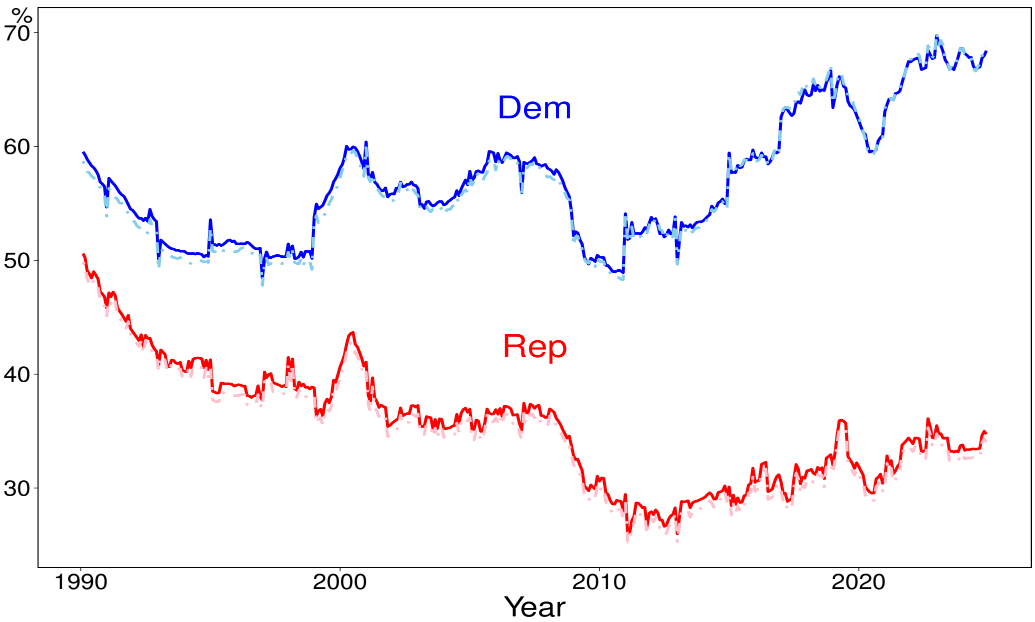


Figure 10: Education level increases by 25% quartile and education diversity decreases by 10% quantile

creasing education level can be even totally offset, as shown in Figure 10. Consequently, the public opinion decreases by 0.34 percentage points (0.73 percentage points) for democratic (republican) counties. This suggests that education investment should be universal, so that the improvement in overall education level is achieved by the development of all education groups. Otherwise, education investment may even backfire and reduce public opinion on climate change.

Existence of only one national television station.

We experiment with a scenario that there is only one national television station in Figure 11 and 12. When only democratic television station is allowed to convey information, the public opinion of republican counties decreases by 0.70 percentage points on average, while that of democratic counties also decreases by 0.21 percentage points. In contrast, when only republican television station exists, public opinion of democratic counties increases by an average of 0.54 percentage points, and that of republican counties also rises slightly by 0.04 percentage points.

At a first glance, one might wonder why the public opinion of counties that hold the same political stance with the only existing national television station also slightly deviates, even though the news they broadcast do not change. This is because of the spatial interaction in our model, meaning that

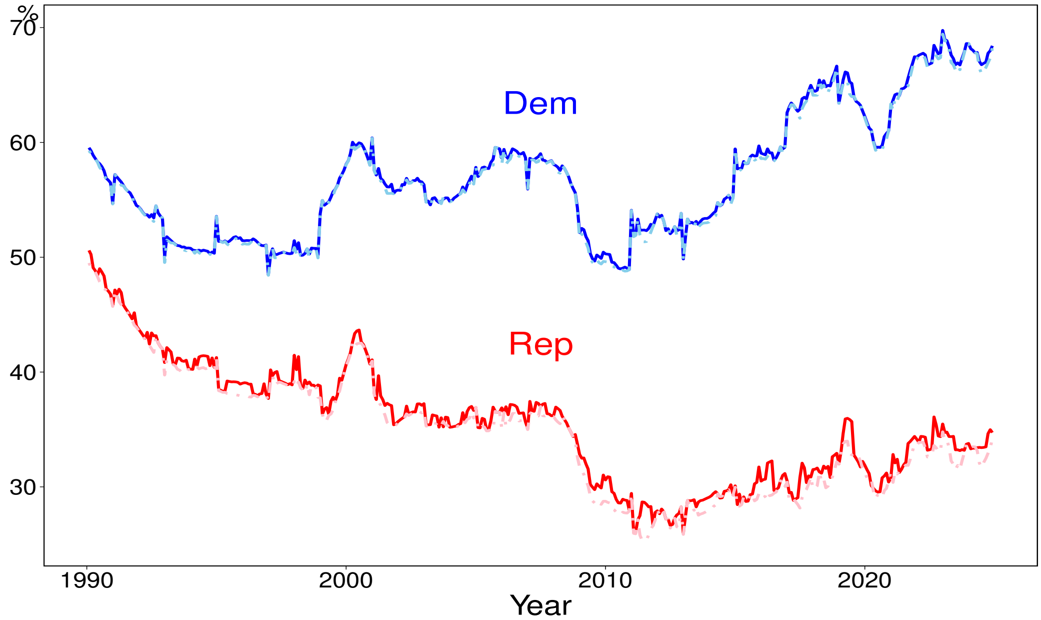


Figure 11: Existence of only democratic television station

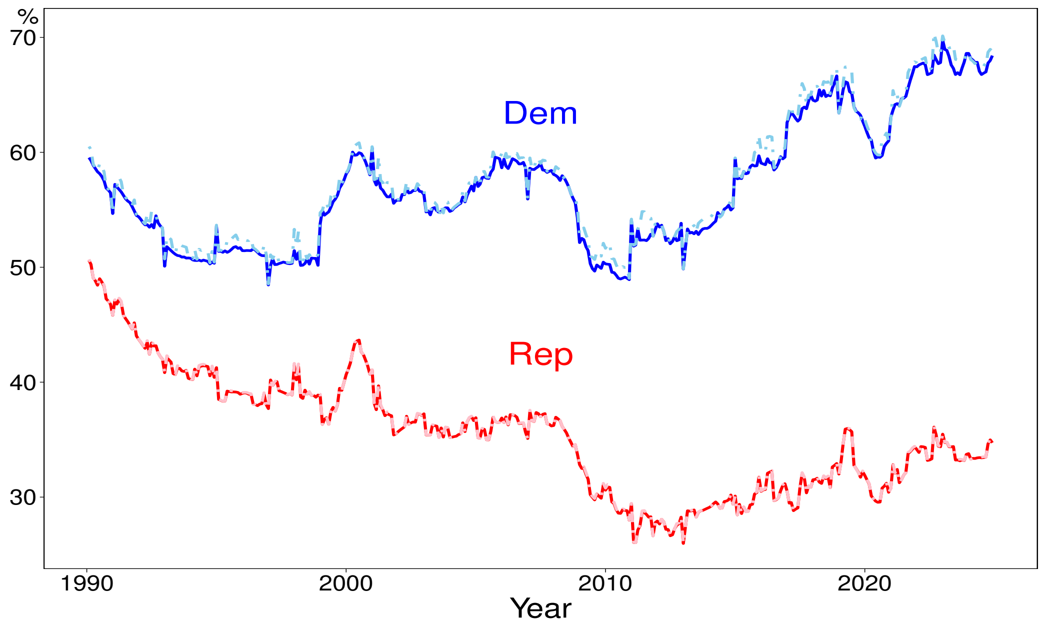


Figure 12: Existence of only republican television station

a change in one county affects all neighboring counties.

Also, one might expect that when individuals are exposed to news from a source with opposing political stance, then they should learn something new from it and thus might change their original opinions. This view is in line with one strand of the literature on media effects, which argues that sustained exposure to opposing media enables people to learn alternative information and thus moderate their attitudes (Broockman and Kalla, 2025). However, there is another widely supported branch of research suggests that cross-cutting media may elicit counterarguments, which in turn reinforces prior attitudes due to the theory of motivated reasoning (Taber and Lodge, 2006; Hart and Nisbet, 2012; Levendusky, 2013). Our experiments results correspond to the latter argument.

Reduce unemployment.

We assume that the monthly unemployment decreases by 25 percentage

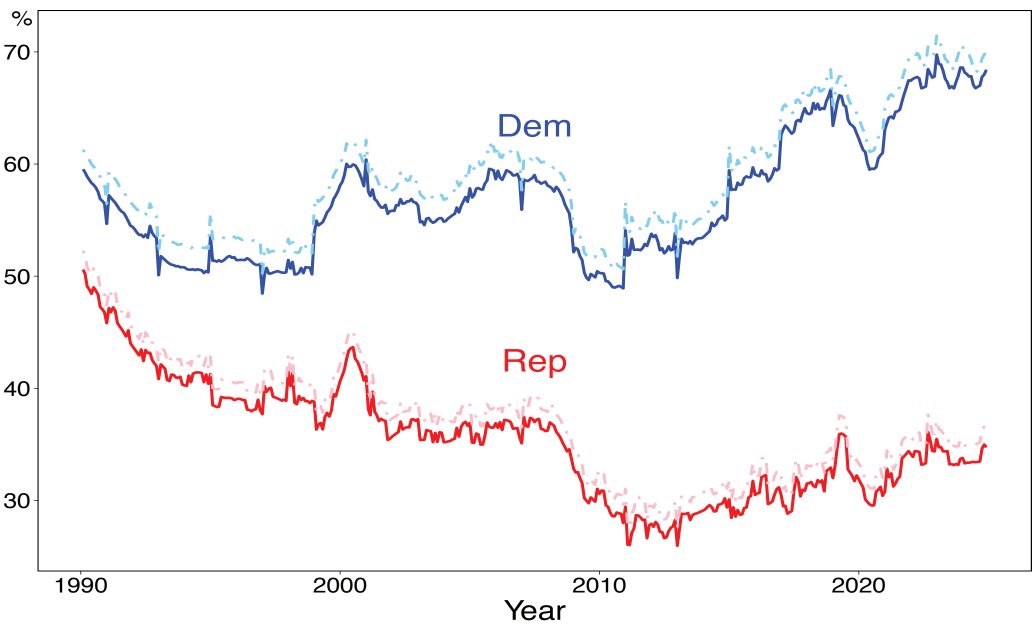


Figure 13: Unemployment decreases by 25% quartile

points (from 50% to 25%). As Figure 13 shows, the public opinion of both democratic and republican counties increase dramatically. Specifically, the opinion of democratic (republican) counties increases by 1.93 percentage point (1.59 percentage point) on average. A reduction in unemployment (economic pressure) allows the public to pay more attention to climate change, and the effect is substantial.

Increase objective positivity and reliability of news items (from

±0.5 ***and 0.5 to*** ±0.9 ***and 0.9).***

In Figure 14, we explore the effects of increasing positivity and reliability

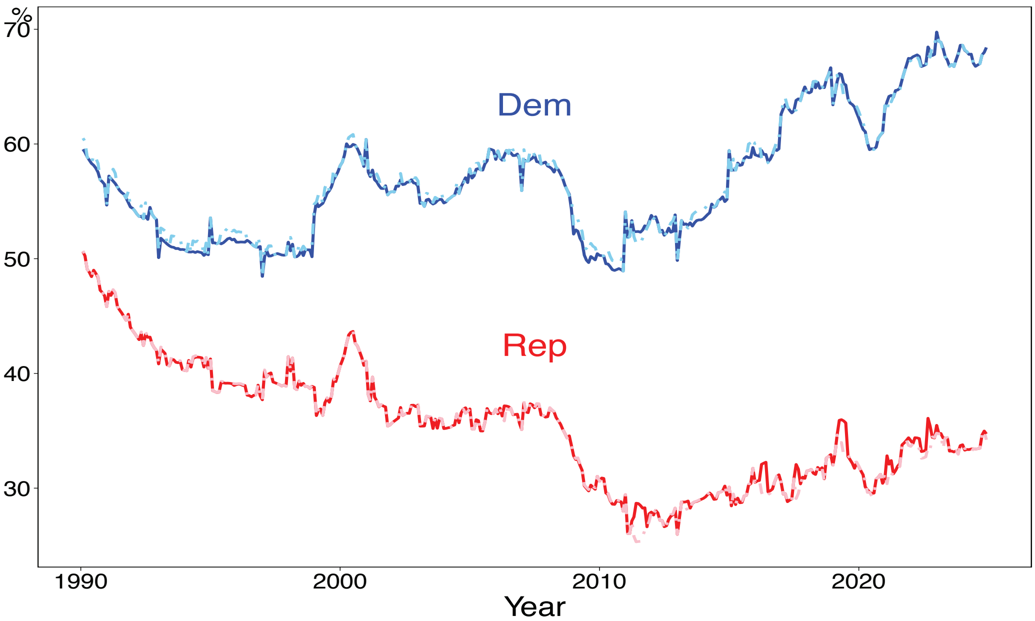


Figure 14: News items with higher positivity and reliability

of news items greatly. According to the figure, the effects vary in direction, sometimes positive and sometimes negative for both groups. However, overall public opinion of democratic counties increases by an average of 0.23 percentage points, and that of republican counties decreases by 0.12 percentage points, which are very small. It implies that news with higher positivity and reliability make democratic counties more concerned about climate change, whereas republican counties become less concerned. Nevertheless, the impacts are extremely limited. The public seems to be indifferent to the positivity and reliability of news.

Adjust news filtering thresholds (Deviation from mean changes from -0.03 ***to*** -0.05 ***for democratic station, from*** $+0.02$ ***to*** -0.02 ***for republican station).***

In Figure 15, we illustrate the effects on public opinion when both national stations lower their thresholds to broadcast more news. It is shown that lower thresholds have relatively large effects on public opinion. The public opinion of democratic (republican) counties decreases by 0.47 percentage points (0.80 percentage points) on average. It seems to be surprise that when more news flow to the public, public concern about climate change actually goes down, for both democratic and republican counties. However, one of the

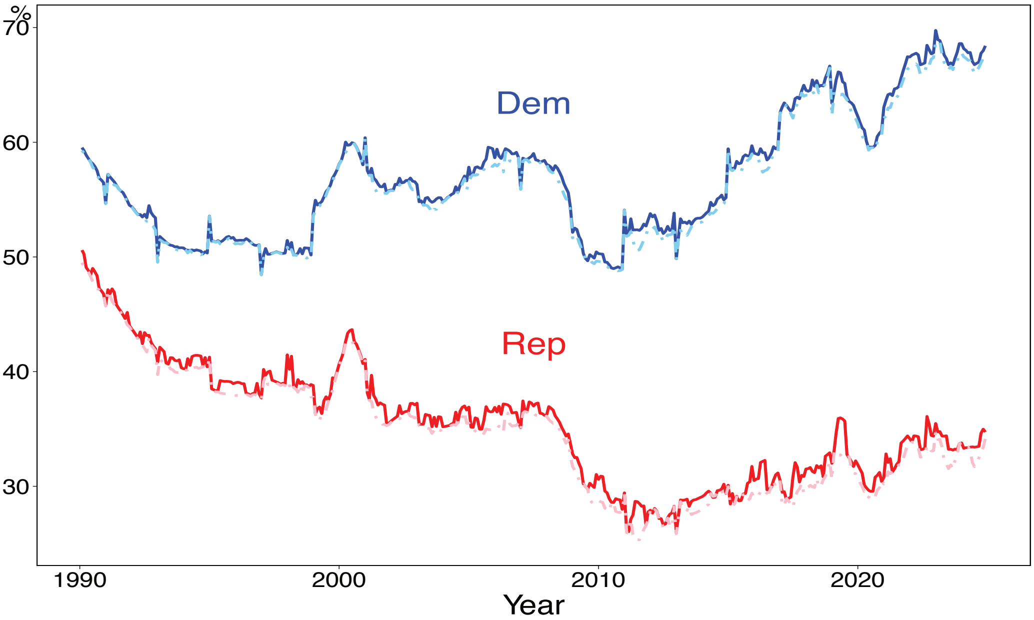


Figure 15: Lower thresholds of national television stations

main distinctions of our study from previous studies is that we quantify the quality of news as well, and include both news importance and news number in the model. Consequently, lower thresholds lead to more news, but at the same time the average importance of news decreases because television stations select more news with relatively lower objective importance than before. This may make people become accustomed to climate change and lower their worries because of the information dilution effect. In fact, the policy experiments potentially reveal the trade-off between news quality and news quantity. It seems that fewer but more important climate-related news can actually push up public opinion about climate change.

Local television stations broadcast news from the national television station holding the opposite political views.

We examine a counterfactual case that individuals are exposed to news from the national station with opposite political stance. Figure 16 indicates that public opinion of democratic counties will slightly increase overall (0.33 percentage points on average), while that of republican counties will decrease moderately (0.66 percentage points on average).

Political distance between national television stations increases.

We investigate the effect on public opinion when national television stations become more extreme in their political stances in Figure 17. As expected, the larger political distance between national television stations intensifies the

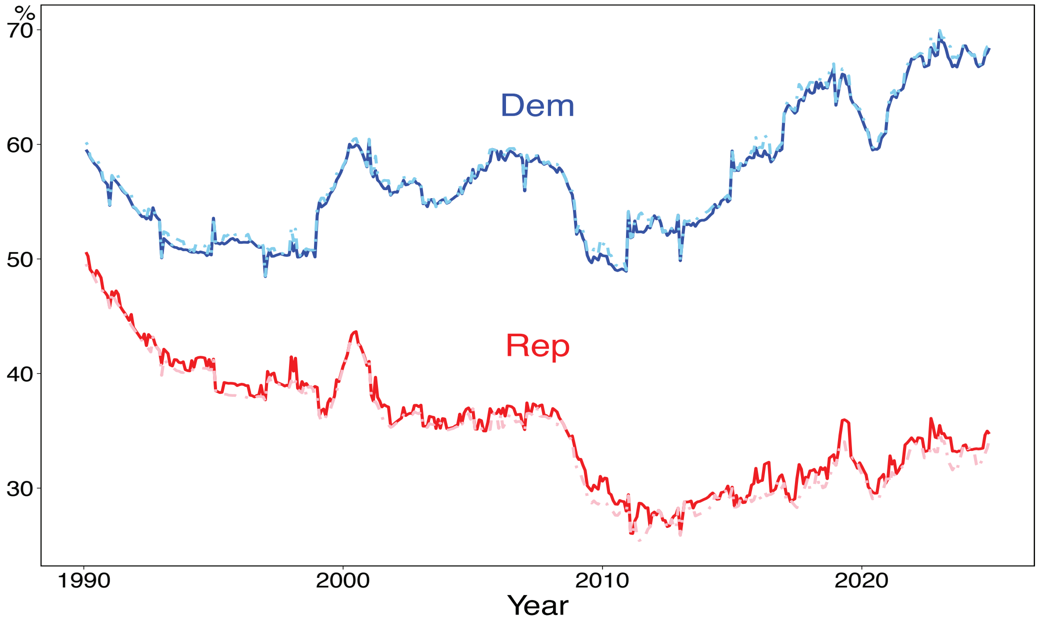


Figure 16: Contrarian behavior of local television stations

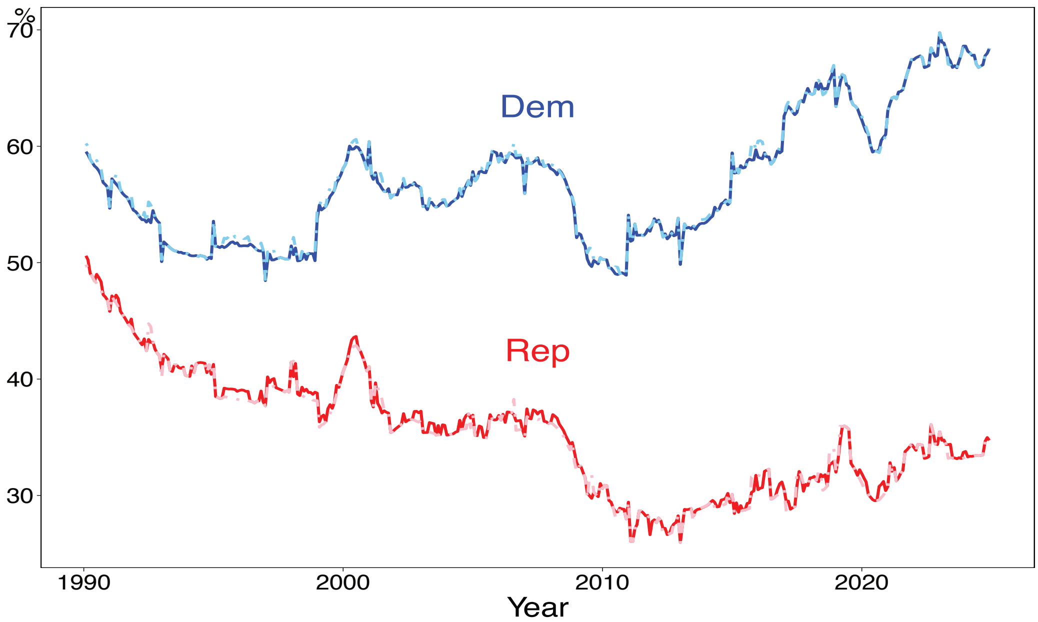


Figure 17: Polarization of two national television stations

polarization in public opinion between two groups of counties (democratic increases by 0.15 percentage points, republican decreases by 0.13 percentage points on average), although the effects are so small that they can be almost neglected.

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